



Chassis Management for NVIDIA Switch Systems

User Manual

Contents

Release Notes Update History	1
Introduction	7
Software Components	7
Host and BMC software stacks	8
Hierarchy and Structure	12
Virtual SysFS Hierarchy	15
Get ASIC Bus	16
Get ASIC I2C Bus	17
Get ASIC PCI Bus	17
Get ASIC Chip-up completed	18
Get ASIC init done	18
Set Chip-down/Chip-up Delay	19
Read CPLD Number	19
Read PSU VPD Info	20
Get Hot-plug Fan Number	20
Get Hot-plug PSU Number	21
Get Hot-plug PDB Number	21
Get Hot-plug PWR Number	22
Get Hot-plug Linecards	22
Read Gearbox Counter	23
Read Module Counter	23
Read ASIC Chipup Counter	24
Read ASIC Chipup Completed	24
Read Init Done	25
Read Max System Fans (rotors)	25
Read Fan Drawer Number	26
Read Fan Command	26
Read Fan Max/Min Speed	27
Read Fan Max/Min Speed for front/rear FAN	27
Fan Speed Tolerance	28

Fan Speed Units	28
Number of Leakage Sensors	29
Number of Cable Cartridges	30
Read/write Time Window for Thermal Control Periodic Log Report	30
Read PSU I2C Address	31
Read PSU I2C Bus	31
Read Thermal Delay	32
Dummy PSUs Supported	32
Read PSU power capacity	33
Read PSU power slope	33
Read DPU Number	34
Read DPU Board type	34
Read DPU board bus offset	35
Read DPU bus offset	35
Read DPU events	36
Read DPU events to host	36
Labels Ready	37
CPU Type	37
Named Busses	38
I2C Bus Offset	38
I2C Bus Connect Devices	39
I2C Bus Default Off EEPROM CPU	39
I2C Comex Mon Bus Default	40
I2C SWB Bus	40
LM Sensors Configuration	41
LM Sensor Labels	41
Events Ready	42
Minimal Driver Unsupported	42
SGMII PHY	43
System Flow Capability	43
Fan Direction EEPROM	44
Global Write Protection Wait Step	44
Global Write Protection Timeout	45
ConnectX Default I2C Bus	45
JTAG Bridge Offset	46
Core 0 Temperature ID	46
Core 1 Temperature ID	47
Read PDB Hotswap Scale Factor	47

Read LED Control Type Map	48
Cable Cartridge FRU Validity	49
BIOS Control	49
BIOS Active Image	50
EEPROM Control	50
Read CPU EEPROM Data	50
Read Fan Module EEPROM Data	51
Read Power Supply Module EEPROM Data	51
Read System Chassis EEPROM Data	52
Read System Chassis EEPROM Parsed Data	52
Read Cable Cartridge EEPROM Data	53
Read system EEPROM (BMC stack)	53
Read BMC board EEPROM (BMC stack)	54
Environment Control	54
Get A2D Voltage	54
Get Comex Voltage Current	55
Get Comex Voltage Input	55
Get Comex Voltage Power	56
Get System Voltage Current	56
Get System Voltage Input	57
Get System Voltage Power	57
Get PDB Hotswap Controller Current	58
Get PDB Hotswap Controller Voltage	58
Get PDB Hotswap Controller Power	59
Get PDB Hotswap Power Scale	59
Get PDB Hotswap Current Scale	60
Get PDB Hotswap Controller Thresholds	60
Get PDB Power Converter Current	61
Get PDB Power Converter Voltage	62
Get PDB Power Converter Power	63
Get PDB Power Converter Thresholds	64
Events	64
Get FAN hot-plug event status	64
Get PSU hot-plug event status	65
PWR hot-plug event status	65
PDB hot-plug event status	66
DPU Ready event	66
DPU Shutdown event	67

EROT (External Root of Trust) AP event	67
EROT (External Root of Trust) Error Event	68
Graceful Power Off Event	68
Power Button Event	69
Power Events	69
Power Button Event	70
Alarms	70
Get PMBUS voltmon alarm status.	70
Get COMEX PMBUS voltmon alarm status.	71
Get PSU PMBUS alarm status.	71
Get PDB Hotswap Controller alarm status	72
Get PDB Power Converter alarm status	73
Get CPU temp alarm status.	73
PSU FW	74
Get Secondary FW version of PSU	74
Get Primary FW version of PSU	74
DPU system attributes	74
Get DPU id	75
Get DPU boot progress	75
Get DPU cpld version	76
Get DPU cpld base version	76
Get DPU auxiliary power reset reason	77
Get DPU main board reset reason	77
Get DPU comex power failure	78
Get DPU power reset reason	78
Get DPU thermal shutdown reason	79
Get DPU tpm reset reason	79
Get DPU perst reset reason	80
Get DPU phy reset reason	80
Get DPU usb phy reset reason	81
Get DPU fpga part number	81
Get DPU fpga minor version	82
Get DPU ufm upgrade status	82
Get DPU VR update status	83
DPU events	83
Get DPU PLL power good indication	83
Get DPU input power indication	84
Get DPU serdes power indication	84

Get DPU serdes analog power indication	85
Get DPU core power indication	85
Get DPU cpu power indication	86
Get DPU digital interfaces power	86
Get DPU ddr5 power indication	87
Get DPU thermal trip indication	87
Get DPU tps upgrade status	88
Get DPU cpu power fault indication	88
Get DPU cpu VR hot alert	89
Get DPU ddr5 fault indication	89
Get DPU ddr5 hot alert	90
LC Alarms	90
Get LC Hot Swap Power Alarm	90
Get LC Voltage Input Alarm	91
Get LC Voltage Current Alarm	91
Get LC Voltage Power Alarm	92
LC EEPROM	92
Read LC EEPROM FRU	92
Read LC EEPROM INI	93
Read LC EEPROM VPD Parsed	93
Read LC EEPROM INI Parsed	94
LC Environment	94
Get LC Voltage Current	94
Get LC Voltage Input	95
Get LC Voltage Power	95
Get LC Hot Swap Current	96
Get LC Hot Swap Input	96
Get LC Hot Swap Power	97
Get LC A2D Voltage	97
Get LC A2D Voltage Scale	98
LC LED	98
Get LC Status LED	98
Get LC Status LED Capabilities	99
Set LC Status Green/Orange	99
Set LC Status LED Green/Orange Delay Off	100
Set LC Status LED Green/Orange Delay On	100
LC Config	100
Read LC CPLD Number	101

Read LC FPGA Number	101
Read LC Gearbox Number	102
Read LC Gearbox Manager Number	102
Read LC Port Number	103
Read LC Module Counter	103
UART	104
ASIC control	104
LC thermal	104
Read LC Gearbox Temperature Input	105
Get LC QSFP/SFP Module Thermal	105
Read Temperature Critical Module	106
Read Temperature Emergency Module	106
Read Temperature Fault Module	107
Read Temperature Input Module	107
LED Control	107
Get Fan Status LED	108
Get Fan LED Capabilities	108
Set Fan LED Green/[Amber/Red]	109
Set Fan LED Green/[Amber/Red] Delay Off	109
Set Fan LED Green/[Amber/Red] Delay On	110
Get PSU LED	110
Get PSU LED Capabilities	111
Set PSU LED Green/[Amber/Red]	111
Set PSU LED Green/[Amber/Red] Delay Off	112
Set PSU LED Green/[Amber/Red] Delay On	112
Get Status LED	113
Get Status LED Capabilities	113
Set Status Green/[Amber/Red]	114
Set Status LED Green/[Amber/Red] Delay Off	114
Set Status LED Green/[Amber/Red] Delay On	115
Get LED Control Owner	115
Power Control	116
Get Power Consumption	116
Get PSU sensor Current + thresholds	117
Get PSU sensor Voltage + thresholds	117
Get PSU sensor Power + thresholds	118
Get PSU sensor capability	118

System / Power Control	118
Get ASIC Health	119
MCU Reset Control	120
Get CPLD Major Version	120
Get CPLD Part Number	121
Get CPLD Minor Version	121
Get CPLD Full Version	122
Fan Direction	122
Set JTAG Mode	123
Set PSU On/Off	123
Set System Power Cycle	124
Set System Power Down	124
CPU Shut Down Request	125
Graceful Power Off	125
CPU Power off Ready	126
Shutdown Unlock	126
Boot Completed	127
Set Line Card Power	127
Set Line Card Enable	128
Read Line Card Active	128
Read Line Card Powered	129
Read Line Card Present	129
Read Line Card Ready	130
Read Line Card Synced	130
Read Line Card Verified	131
Read Line Card Reset Mask	131
Set Line Card Shutdown	132
Set VPD Write Protect	132
Set ASIC Up during PCIe root complex reset	133
Get Voltreg Update status	133
Get Config1, Config2	134
Get Ufm Version	134
DPU Power Off	135
DPU Force Power Off	135
DPU Reset	136
DPU Reset Enable	136
Get Reset Cause	137
Cartridge	138

ASIC PG Failure	139
Leakage Sensors	139
Leakage Sensor	140
Thermal	140
Ambient sensors	141
Read Switch ASIC Temperature	141
Read Switch ASIC Temperature Normal	142
Read Switch ASIC Temperature Critical	142
Read Switch ASIC Temperature Emergency	143
Read Switch ASIC Temperature Trip Critical	143
Read Switch Comex Temperature	144
MNG Temperature	144
Read host CPU temperature (BMC stack)	145
Read host CPU temperature max / min (BMC stack)	145
Read BMC Temperature	146
Read BMC Max Temperature	147
Read BMC Critical Temperature	148
Read BMC Minimal Temperature	148
Read Cooling State	149
Cooling Name	149
Read CPU Core Temperature	150
CPU Core Critical Temperature	150
CPU Core Critical Temperature Alarm	151
CPU Core Temperature Max	151
Read CPU Pack Temperature	152
CPU Pack Critical Temperature	152
CPU Pack Critical Temperature Alarm	153
CPU Pack Temperature Max	153
Read Fan Max Speed	154
Read Fan Min Speed	154
Set Fan Speed	155
Fan Speed Tolerance	155
Read Fan Direction	156
Read Fan Status	156
Read Fan Fault	157
Comex Voltmon Temperature	157
Comex Voltmon Critical Temperature	158
Comex Voltmon Max Temperature	158

Read Port Ambient	159
Read PSU Temperature	159
Read PSU Alarm	160
Read PSU Max	160
Read PSU Fan Speed	161
Read PSU min/max Fan Speed	161
Read PSU Power Status	162
Read PSU Status	162
Read PDB Hotswap Controller Temperature	163
Read PDB Hotswap Controller Temperature Thresholds	163
Read PDB Power Converter Temperature	164
Read PDB Power Converter Temperature Thresholds	165
Read PDB MOSFET Ambient Temperature	165
Read System PWM1	166
Read Temperature Critical Module	166
Read Temperature Emergency Module	167
Read Temperature Trip Critical Module	167
Read Temperature Fault Module	168
Read Temperature Input Module	168
Read Temperature Critical Gearbox	169
Read Temperature Emergency Gearbox	169
Read Temperature Trip Critical Gearbox	170
Read Temperature Input Gearbox	170
Read Switch CPU Temperature	171
Read Switch Fan Temperature	171
Read Switch Port Temperature	172
Read Switch Power Supply Temperature	172
SODIMM Temperature Input	173
SODIMM Critical Temperature	174
SODIMM Max Temperature	174
SODIMM Min Temperature	175
SODIMM Temperature Alarms	176
SODIMM Temperature Hysteresis	177
SWB ASIC Temperature	177
Drive Temperature	178
Drive Critical Temperature	178
Drive Maximum Temperature Threshold	178
Drive Minimum Temperature Threshold	179

Watchdog	179
Read Boot Status	179
Read Identity	180
Read No Way Out	180
Read State	181
Read Status	181
Read Timeout	182
Read Timeleft	182
BMC status	182
BMC present	183
BMC to CPU control	183
MCTP address	184
MCTP bus	184
MCTP ready	185
BMC reset cause	185
Primary BMC reset cause	185
BMC reset domain detail	186
Raw SCU reset event logs	187
BMC leakage A2D tree	188
Leak detector attributes	188
Leak channel attributes	189
JTAG interface	190
Enable / Disable JTAG mechanism	190
Set JTAG TCK pin	191
Set JTAG TDI pin	191
Set JTAG TMS pin	192
Get JTAG TDO pin	192
Thermal Control	192
Thermal Control Versions	193
TC v2.0 (Default)	193
TC v2.5 (Advanced)	193
Drivers	193
Watchdog	193

Release Notes Update History

Revision	Date	Description
3.2.8	August 2026	§3.1.61 config/cable_cartridge<n>_valid: IPMI FRU check result per cable cartridge (N51XX_LD / N61XX_LD)
3.2.7	August 2026	§3.1.60 config/led_control_type platform LED owner map; §3.16.28 led/led_<name>_control (led_sw / led_hw / led_hw_sw); glob masks * and ?
3.2.6	June 2026	§3.1.59 and §3.4 PDB hotswap scale (SN6600_LD / lm5066i); §3.24 AST2700 BMC reset cause tree; §3.25 BMC A2D leakage runtime layout; N6300_LD (HI185) platform notes; §3.18 cpu_shutdown_req hw-mgmt polling note
3.2.5	June 2026	§2.2 HI189 BMC peripheral table; §3.3.7–§3.3.8 BMC EEPROM bodies; §3.20 BMC stack tags and CPU-on-BMC thermal cross-refs; §3.23 BMC status bodies (present, bmc_to_cpu_ctrl, MCTP)
3.2.4	June 2026	Added §2.2 Host and BMC software stacks : separate repo paths (usr/ vs bmc/usr/), packages, handlers, and examples; stack notes in §3 intro and §3.20 Thermal

Revision	Date	Description
3.2.3	June 2026	§3.20 thermal: filled missing section bodies; TOC aligned; per-HID BMC examples under <code>bmc/examples/<HID>/examples/</code>
3.2.2	June 2026	BMC thermal sysfs (HI189 / 1m75): documented <code>bmc_temp_input</code> and <code>bmc_temp</code> ; removed obsolete TOC entries for <code>bmc_crit</code> / <code>bmc_min</code> (not created on BMC stack)
3.2.1	May 2026	Corrected N51XX_LD platform family reset-cause list in Get Reset Cause (#5014001)• Documented 22 CPLD-supported reset causes for N51XX_LD/GB200 systems• Removed unsupported causes: <code>reset_ac_pwr_fail</code> , <code>reset_aux_pwr_or_ref</code> , <code>reset_from_asic</code> , <code>reset_reload_bios</code>

Revision	Date	Description
3.2	March 2026	Added SN6600_LD (SN66XX_LD family, SKU HI193) liquid-cooled platform support• Validation: <code>usr/etc/hw-management-sensors/sn66xxld_sensors.conf</code>, <code>hw-management.sh</code> / event scripts Platform notes:• Single ASIC (<code>asic_num=1</code>), 4 CPLDs, <code>hotplug_pdb=2</code>, <code>pdb_hotswap1/2</code> and <code>pdb_pwr_conv1/2</code>• ASIC voltmons: 19 sysfs indexes (<code>voltmon1-14</code>, <code>voltmon16-20</code> on captured tree)• SODIMM temp: JC42 at 0x52/0x53 on I2C bus 10• Watchdog: <code>watchdog/main/</code> and <code>watchdog/aux/</code> hierarchy• PDB hot-plug events: <code>events/pdb1</code>, <code>events/pdb2</code> Updated Sections:• Liquid-cooled applicability notes extended to SN66XX_LD across environment, alarms, thermal, and leakage-related text• Config: documented optional <code>psu<X>_i2c_bus</code> (hw-management internal; OS must not require it)

Revision	Date	Description
3.1	January 2026	Added N6100_LD (N61XX_LD family) liquid-cooled multi-ASIC platform support New Sections for N6100_LD: • Multi-ASIC Health (asic_health, asic2_health, asic3_health, asic4_health) • MCU Reset Control (mcu1_reset, mcu2_reset) • Cable Cartridge EEPROM (cable_cartridge1-4_eeprom) • Cartridge Counter (config/cartridge_counter) • Cartridge Status (cartridge1-4) • eRoT Events (erot1_ap, erot1_error) • Config: asic_num=4, erot_count=1 Updated Sections: • Power Converters: Added pwr_conv naming (vs pdb_pwr_conv for SN58XX_LD) • Updated all liquid-cooled references to include N61XX_LD family • Extended voltmon support for 16 PMICs (voltmon1-16) • SODIMM Temperature Sensors: Updated to include both SN58XX_LD and N61XX_LD

Revision	Date	Description
3.0	September 2025	Complete document alignment with Word document source• Updated title and branding to NVIDIA• Complete sysfs hierarchy coverage with 300+ attributes• Professional markdown formatting throughout• Added comprehensive examples for all attributes• Updated all 22 major sections (3.1-3.22)• Added Watchdog, JTAG, and BMC sections• Complete thermal monitoring documentation• Enterprise-grade documentation ready for production
2.8	April 1st 2024	Added temperature, BMC and power related attributes
2.6	July 1st 2024	Added DPU related attributes
2.4	Aug 31, 2023	Adding 'asics_init_done' and 'asic_chipup_completed'
2.3	July 11, 2023	Update LEDs colors to be either red or amber for FAN LED, PSU LED and status LED
2.2	Feb 15, 2022	Add many SN4800 related attributes Add PSU FW version related attributes
2.1	Sept 15, 2021	Add PSU MIN/MAX fan speed. Added the following sections:• Get psu sensors value.• Get psu sensors thresholds.• Get psu sensors capability.
2.0	May 25, 2021	Edit reset causes - page 31-32 Add spectrum 3Remove comex_wd reason which is disabled.

Revision	Date	Description
1.9	Dec 30, 2020	Added updates for Fan Direction JTAG
1.8	July 01, 2020	Added the following sections:• Read PSU VPD Info• Get Hot-plug Fan Number• Get Hot-plug PSU Number• Get Hot-plug PWR Number• Get FAN hot-plug event status• Get PSU hot-plug event status• PWR hot-plug event status• Read PSU min/max Fan Speed• Read/write Time Window for Thermal Control Periodic Log Report
1.7	Apr 13, 2020	Added the following sections:• 2.2.3 Read SFP Counter• 2.2.4 Read Module Counter• 2.2.5 Read Max System Fans (rotors)• 2.2.6 Read Fan Drawer Number• 2.6.3 Get CPLD Part Number• 2.6.4 Get CPLD Minor Version• 2.6.5 Get CPLD Full VersionModified the following sections:• 2.3.2 Read Fan Module EEPROM Data• 2.6.2 Get CPLD Major Version• 2.7.19 Read PSU Temperature• 2.7.26 Read Temperature Critical Module• 2.7.27 Read Temperature Emergency Module• 2.7.28 Read Temperature Fault Module• 2.7.29 Read Temperature Input Module
1.6	Apr 12, 2020	Modified “2.6.8 Get Reset Cause”
1.5	Nov 27, 2019	Modified “2.6.8 Get Reset Cause”
1.4	Sept 23, 2019	Added “2.6.3 Fan_Dir”Modified “2.6.8 Get Reset Cause”
1.3	June 13, 2019	Added:• Thermal• Watchdog

Revision	Date	Description
1.2	April 12, 2019	Updated Sysfs
1.1	December 18, 2018	Added support for new systems
1.0	September 8, 2015	First release

Introduction

NVIDIA hw-management package uses a virtual file system provided by the Linux kernel called sysfs.

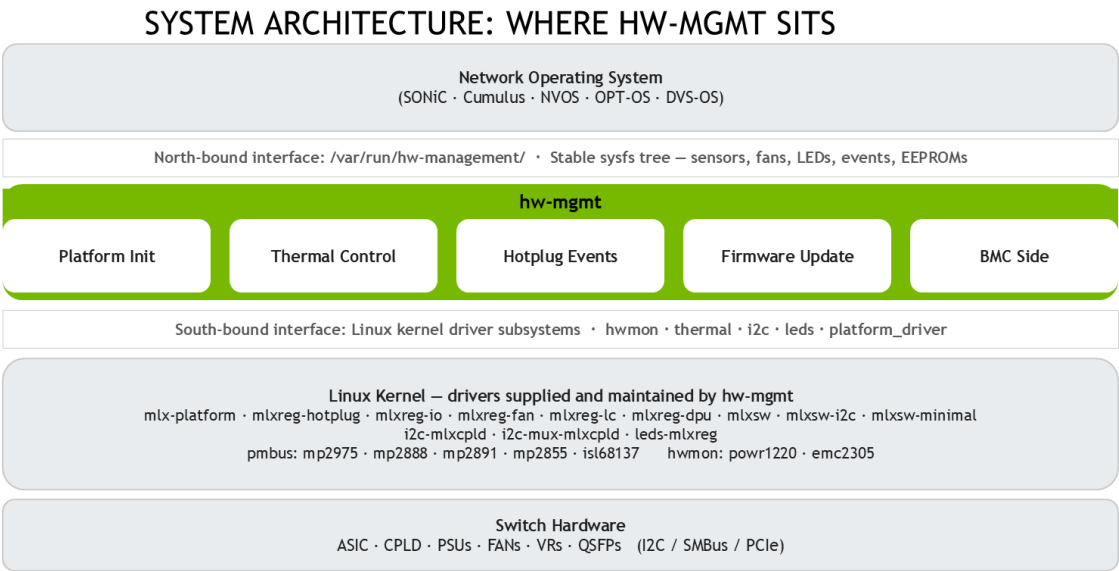
The sysfs file system enumerates the devices and buses attached to the system in a file system hierarchy that can be accessed from the user space.

The major advantage of working with sysfs is that it makes HW hierarchy easy to understand and control without having to learn about HW component location and the buses through which they are connected.

Software Components

Figure 1 presents the system architecture and layer separation for sysfs support.

Figure 1 - Sysfs Layout



Host and BMC software stacks

The **hw-mgmt** repository ships two Debian packages that both expose the same virtual hierarchy under `/var/run/hw-management/` (`$bsp_path` in this manual), but they are built from **different source trees** and run on **different processors**:

	Host stack	BMC stack
Role	Switch CPU / NOS (SONiC, ONL, Cumulus, ...)	Dedicated BMC SoC (for example AST2700 on Microsoft Sonic BMC OS)
Debian package	hw-management	hw-management-bmc
Repository source	usr/ (scripts under usr/usr/bin/, platform data under usr/etc/)	bmc/usr/ (mirrors host layout; platform data under bmc/usr/etc/<HID>/)

	Host stack	BMC stack
Installed on target	/usr/bin/hw-management*.sh, /usr/bin/hw_management_thermal_control*.py, /lib/udev/rules.d/50-hw-management-events.rules, ...	/usr/bin/hw-management-bmc*.sh, /lib/udev/rules.d/5-hw-management-bmc-events.rules, /etc/<HID>/, ...
Primary event handlers	hw-management-chassis-events.sh, hw-management-thermal-events.sh	hw-management-bmc-events.sh (and helpers under the same prefix)
Init / systemd	hw-management.service	hw-management-bmc-init.service and related BMC units (see bmc/README.md)
Examples / reference layouts	Platform sensors in usr/etc/hw-management-sensors/; scripts under usr/usr/bin/	Flat files under bmc/examples/; per-HID template at bmc/examples/HIxxx/examples/
Developer guide	Repository root README.md	bmc/README.md, bmc/DEVELOPER_GUIDE.md

How to use this manual

- **\$3.x attribute sections** describe `$bsp_path nodes` as seen by applications on either CPU or BMC when that node is created on the platform.
- Sections that name **host-only** handlers (for example `hw-management-thermal-events.sh`, `hw_management_thermal_control.py`) apply to the **host package** unless stated otherwise.
- Sections that name `hw-management-bmc-*` or **HI189 / BMC thermal stack** apply to the **BMC package** on systems that ship it.
- Some nodes exist on **one stack only** (for example host ASIC/voltmon thermal vs BMC `bmc_temp_input` from `hw-management-bmc-events.sh`). Check the platform or the validated system tree for your SKU.

Both stacks may be present on the same product (CPU + BMC each running their own package); they do **not** share the same `/usr/bin` install tree on a single root filesystem.

Stack column (used in §3.x):

Label	Meaning
Host	Node created by <code>hw-management</code> on the switch CPU (<code>hw-management-thermal-events.sh</code> , <code>hw-management-chassis-events.sh</code> , <code>hw-management.sh</code>)
BMC	Node created by <code>hw-management-bmc</code> on the BMC SoC (<code>hw-management-bmc-events.sh</code> and related BMC scripts)
Host + BMC	Same logical name may exist on both images; paths/handlers differ — see section note

BMC stack — HI189 reference (`hw-management-bmc-events.sh`, **validated against** `bmc/usr/etc/HI189/5-hw-management-bmc-events.rules` **and** `hw-management-bmc-early-i2c-devices.json`):

\$bsp_path subtree	Symlinks / behaviour	Peripheral / driver	Stack
<code>thermal/cpu_temp_input</code> , <code>cpu_temp</code> , <code>cpu_min</code>	Host CPU temperature via BMC-side I2C	sbtsi	BMC
<code>thermal/bmc_temp_input</code> , <code>bmc_temp</code>	BMC board ambient (<code>bmc_min</code> not created — lm75 has no <code>temp1_min</code>)	lm75	BMC
<code>eeeprom/eeeprom_system</code>	System VPD EEPROM	24c512	BMC
<code>eeeprom/eeeprom_bmc</code>	BMC FRU EEPROM	24c02	BMC
<code>system/*</code> (many attrs)	mlxreg-io / mlxreg-hotplug register files	CPLD / platform control on BMC	BMC
<code>leakage/<N>/<j>/...</code>	A2D leak-detector tree (input, thresholds, type)	ADS1015 / ADS7924 / MAX1363 per JSON config	BMC

\$bsp_path subtree	Symlinks / behaviour	Peripheral / driver	Stack
bmc/reset_*, bmc/ domains/reset_*, bmc/ raw_scu*	AST2700 BMC reset-cause exporter	hw-management-bmc-get- reset-cause.sh	BMC

Example layouts: `bmc/examples/hw-management-bmc-thermal-sysfs.txt` (delivered path), `bmc/examples/hw-management-bmc-EEPROM-sysfs.txt`, `bmc/examples/hw-management-bmc-system-sysfs.txt`, `bmc/examples/hw-management-bmc-leakage-sysfs.txt`.

Host stack — BMC-related nodes (CPU image, not the BMC package):

Node	Implementation	Stack
system/bmc_ present	GPIO symlink (host <code>hw-management.sh</code> platform GPIO mapping)	Host
system/bmc_to_ cpu_ctrl	mlxreg-io on CPU (mlxplat; same tree)	Host
system/cpu_ mctp_ready	mlxreg-io on CPU (same tree)	Host
config/mctp_ addr, config/ mctp_bus	Written by <code>hw-management.sh</code> when platform defines MCTP (e.g. N5110)	Host

Do not assume host ASIC/PSU/fan thermal nodes exist on the BMC image, or that BMC `cpu_temp_*` nodes exist on the CPU image — check `$bsp_path` on the target root filesystem.

Virtual / SimX platforms (no production sysfs tree in repo): HI194 (N7200_LD), HI199 (N7300_LD), HI200 (N7400_LD) and other SimX SKUs may exit early from `hw-management-ready.sh` with mock values only (`check_simx` paths in `hw-management.sh`). Do not use these for production attribute validation.

Hierarchy and Structure

The package uses the Linux default hierarchy structure of sysfs under the directory `/var/run/hw-management`.

This path is used by existing applications that use auto-discovery to find existing HW components. Two examples for such applications are:

- `libsysfs` – the libraries provide a consistent and stable interface for querying system device information exposed through sysfs.
- `systool` – a utility built upon `libsysfs` that lists devices by bus, class, and topology.

The disadvantage of using this path is that the hierarchy model includes the BUS type and location model which is subject to change between different system types.

To resolve this limitation, the virtual hierarchy structure that is not HW dependent is supported. This hierarchy is a collection of soft links to the default sysfs structure. This document describes the way to work with this hierarchy in order to control the HW.

Chassis attributes information exported through sysfs can be utilized by a number of standard Linux tools. So, for example, the following are tools from the Linux packages `lm-sensors` and `fancontrol`, which are capable of operating on top of sysfs infrastructure:

- `pwmconfig` – tests the pulse width modulation (PWM) outputs of sensors and configures `fancontrol`
- `fancontrol` – automated software-based fan speed regulation
- `sensors` – print sensors information

As described in the previous sections, sysfs structure provides access to HW drivers. These drivers need to be initialized before using sysfs. In addition, NVIDIA virtual hierarchy also needs to be created in order to use it.

The following applies to the **host** package (`hw-management`). For the **BMC** package, see §2.2 and `bmc/README.md` (systemd units, udev rules, and scripts under `hw-management-bmc-*`).

The package provides a simple way to initialize the drivers using the set of shell scripts. These scripts support initialization and de-initialization of driver, virtual hierarchy structure, udev events handling, based on a set of NVIDIA system specific udev rules.

Package contains the following files, used within the workload:

- `/lib/systemd/system/hw-management.service`: system entries for thermal control activation and de-activation.
- `/lib/udev/rules.d/50-hw-management-events.rules`: udev rules defining the triggers on which events should be handled. When trigger is matched, rule data is to be passed to the event handler (see below file `/usr/bin/hw-management-events.sh`).
- `/usr/bin/hw_management_thermal_control.py`: contains thermal algorithm implementation (TC v2.0). `/usr/bin/hw_management_thermal_control_2_5.py`: contains thermal algorithm implementation (TC v2.5).
- `/usr/bin/hw-management-chassis-events.sh` and `/usr/bin/hw-management-thermal-events.sh`: handle udev triggers, according to the received data, it creates or destroys symbolic links to sysfs entries. It allows to create system independent entries, and it allows thermal controls to work over this system independent model. Raises signal to thermal control service in case of fast temperature decreasing. It could happen in case one or few very hot port cables have been removed. Sets PS units internal FAN speed to default value when unit is connected to power source.
- `/usr/bin/hw-management.sh`: performs initialization and de-initialization, detects the system type, connects thermal drivers according to the system topology, activates and deactivates thermal algorithm.
- `/usr/bin/hw-management-led-state-conversion.sh` and `/usr/bin/hw-management-power-helper.sh`: helper scripts.
- `/etc/modprobe.d/hw-management.conf` and `/etc/modules-load.d/hw-management-modules.conf`: configuration for kernel modules loading.

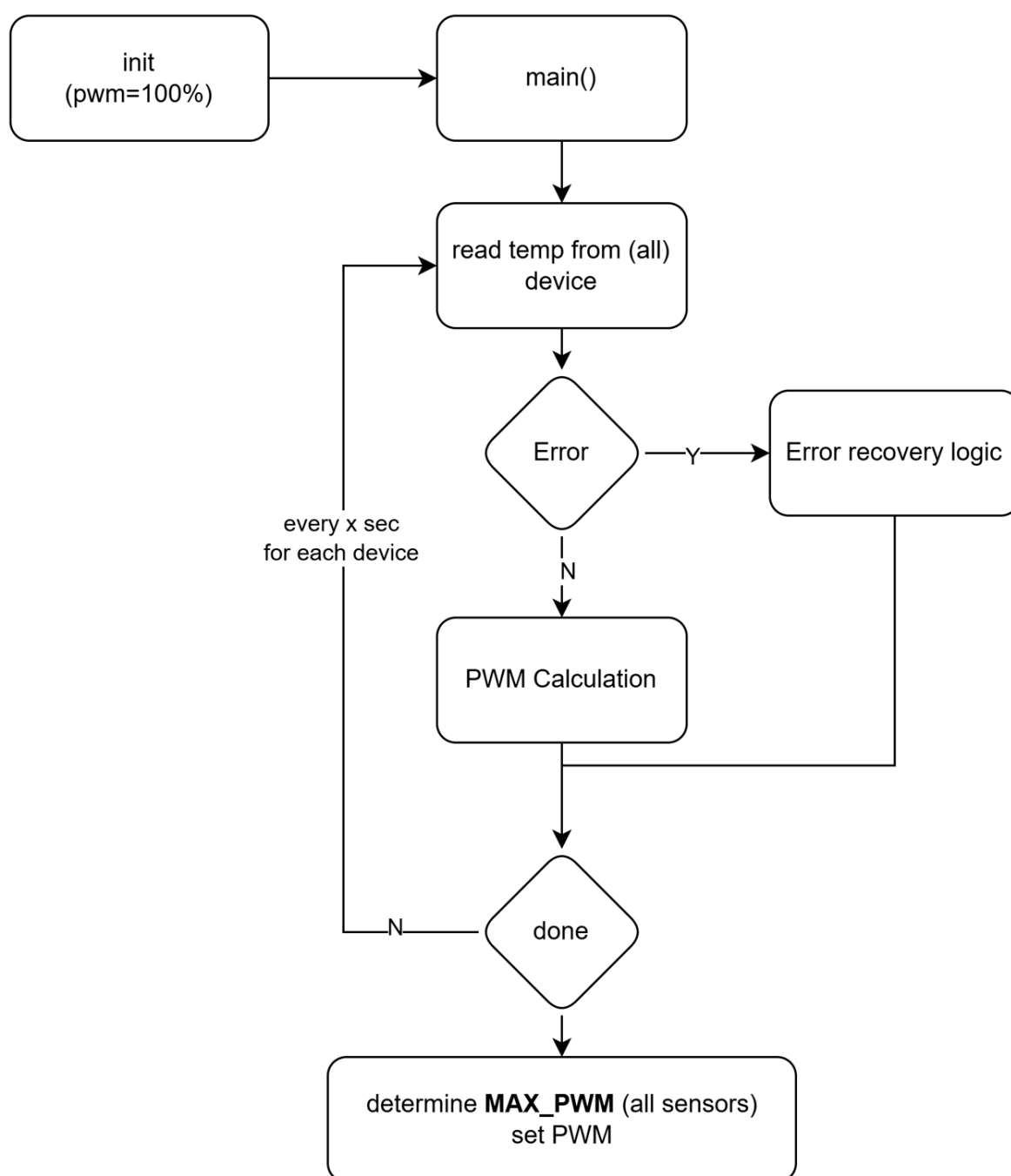
For more details follow package README file (`README.md` for host, `bmc/README.md` for BMC).

Thermal management overview

Thermal control is split into two cooperating layers. A data-collection layer (driven by udev events and periodic polling) reads temperature, fan, and power sensors and publishes them through the sysfs hierarchy described in this manual. A thermal control layer then reads that sysfs data, calculates the required fan PWM (duty cycle) per component, and applies the highest PWM demanded by any sensor to keep every monitored component within its safe operating range. Two thermal control implementations exist — TC v2.0 and TC v2.5 — selected automatically based on the hardware platform; see **Thermal Control** later in this

manual for version details, and the companion Thermal Monitoring for NVIDIA Systems with Third Party OS document for the full algorithm description.

Figure 2 - Thermal Management Flow



Virtual SysFS Hierarchy

NVIDIA virtual hierarchy supports the following HW control (\$bsp_path below is a location of virtual SysFS hierarchy, in standard Linux distributions, like Debian, RedHat, Fedora, etcetera this is /var/run/hw-management folder).

Table 1 - NVIDIA Hierarchy Node Support

Node Path	Purpose
\$bsp_path/config	Internal system specific configuration data
\$bsp_path/eeprom	Gets raw data from EEPROM in system modules
\$bsp_path/environment	Gets information on environmental sensors (A2D, Volt, Curr)
\$bsp_path/led	Gets/sets LED color (see LED_Control_API.md for runtime led_<func> naming and CPLD behaviour)
\$bsp_path/bin	
\$bsp_path/events	
\$bsp_path/firmware	
\$bsp_path/ui	Textual labels for sensors
\$bsp_path/power	Gets information from power sensors
\$bsp_path/system	Gets/sets system variables and settings (CPLD version, fan dir, reset, pwr cycle)
\$bsp_path/thermal	Gets variant thermal sensors in systems and gets/sets fan attributes

Node Path	Purpose
\$bsp_path/watchdog	Standard watchdog sysfs attributes
\$bsp_path/alarm	Get System chassis
\$bsp_path/jtag	Provides interface for JTAG CPLD burn
\$bsp_path/sysfs_labels_rdy	
\$bsp_path/fast_sysfs_labels_rdy	

Detailed information on each of these nodes can be found in the following sections.

Stack applicability: Most §3.x nodes are created by the **host** stack (`hw-management-thermal-events.sh` / `hw-management-chassis-events.sh`). Sections tagged **Stack: BMC** or naming `hw-management-bmc-events.sh` apply to the **BMC package** only (see §2.2). §3.23 documents host-visible BMC status; BMC system/ register layout is in `bmc/examples/hw-management-bmc-system-sysfs.txt`.

Note: some of the attributes described below are not relevant to all platforms and will exist only on the platforms which support this attribute. ## Config Control

Get ASIC Bus

Node name	\$bsp_path/config/asic_bus		
Description	Get system ASIC bus number		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer	1-99
Example	Get asic bus number: <code>cat \$bsp_path/config/asic_bus</code>		

Get ASIC I2C Bus

Node name	\$bsp_path/config/asic<index>_i2c_bus_id		
Description	Get ASIC <index> I2C bus		
Access	Read only		
Release version	V.7.0030.2930		
Arguments	Name	Data type	Values
Example	Get ASIC I2C bus address: cat \$bsp_path/config/asic1_i2c_bus_id		

Get ASIC PCI Bus

Node name	\$bsp_path/config/asic<index>_pci_bus_id		
Description	Get ASIC PCI bus ID		
Access	Read only		
Release version	V.7.0020.1338		
Arguments	Name	Data type	Values
Example	Get asic PCI bus number: cat \$bsp_path/config/asic1_pci_bus_id		

Get ASIC Chip-up completed

Node name	\$bsp_path/config/asic_chipup_completed		
Description	Get ASIC count which has already initialized. When asic_chipup_completed == asic_num, asics_init_done should be set to "1"		
Access	Read		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer	0..asic_count
Example	Get already initialized asic count: cat \$bsp_path/config/asic_chipup_completed		

Get ASIC init done

Node name	\$bsp_path/config/asics_init_done		
Description	Get ASIC init done status. 1 – All asic initialized and ready. 0 – One or more ASICs not ready When asic_chipup_completed == asic_num, asics_init_done should be set to "1"		
Access	Read		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer	1 – ALL ASICs ready 0 – One or more ASICs not ready
Example	Get asics init ready: cat \$bsp_path/config/asics_init_done		

Set Chip-down/Chip-up Delay

Node name	\$bsp_path/config/chipdown_delay \$bsp_path/config/chipup_delay		
Description	Set delay duration in seconds for hw mgmt service from the chip down/up event.		
Access	Write/Read		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer (seconds)	0 – no delay other – delay
Example	Get chipdown value: cat \$bsp_path/config/chipdown_delay Set 5 seconds delay in chipup value: echo 5 > \$bsp_path/config/chipup_delay		

Read CPLD Number

Node name	\$bsp_path/config/cpld_num		
Description	Get the number of CPLD modules in the system		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer	1-X
Example	Get CPLD number: cat \$bsp_path/config/cpld_num		

Read PSU VPD Info

Node name	\$bsp_path/eprom/psu{n}_vpd		
Description	Get PSU VPD info in human readable format		
Access	Read only		
Release version	V.7.0010.1300		
Arguments	Name	Data type	Values
	Status	ASCII	EEPROM info
Example	Get PSU VPD info: cat \$bsp_path/eprom/psu{n}_vpd		

Get Hot-plug Fan Number

Node name	\$bsp_path/config/hotplug_fans		
Description	Get hot-plug FAN number in the system		
Access	Read only It can be zero on fixed system.		
Release version	V.7.0010.1300		
Arguments	Name	Data type	Values
	Status	Integer	0-X
Example	Get hot-plug fan number: cat \$bsp_path/config/hotplug_fans		

Get Hot-plug PSU Number

Node name	\$bsp_path/config/hotplug_psus		
Description	Get hot-plug PSU number in the system. It can be zero on fixed system.		
Access	Read only		
Release version	V.7.0010.1300		
Arguments	Name	Data type	Values
	Status	Integer	0-X
Example	Get hot-plug psu number: cat \$bsp_path/config/hotplug_psus		

Get Hot-plug PDB Number

Node name	\$bsp_path/config/hotplug_pdbns		
Description	Get the number of hot-pluggable Power Distribution Boards (PDB) in the system. Note: This attribute is primarily for liquid-cooled systems (SN58XX_LD family: SN5810_LD, SN5800_LD; N61XX_LD family: N6100_LD, N6300_LD; SN66XX_LD family: SN6600_LD). PDBs manage power distribution in liquid-cooled systems where traditional PSUs are not present. It can be zero on air-cooled systems or systems without hot-pluggable PDBs. Note: N6100_LD has hotplug_pdbns=0 (PDB is not hot-pluggable). Note: N6300_LD (SKU HI185) has hotplug_pdbns=2. Note: SN6600_LD (SKU HI193) has hotplug_pdbns=2 with events/pdb1 and events/pdb2.		
Access	Read only		
Release version	V.7.0040.3930		
Arguments	Name	Data type	Values
	Status	Integer	0-X (number of hot-pluggable PDBs)
Example	Get hot-plug PDB number: cat \$bsp_path/config/hotplug_pdbns		

Get Hot-plug PWR Number

Node name	\$bsp_path/config/hotplug_pwrs		
Description	Get hot-plug Power cable number in the system. It can be zero on fixed system.		
Access	Read only		
Release version	V.7.0010.1300		
Arguments	Name	Data type	Values
	Status	Integer	0-X

Get Hot-plug Linecards

Node name	\$bsp_path/config/hotplug_linecards		
Description	Get the number of Linecards		
Access			
Release version			
Arguments	Name	Data type	Values
Example	Get the number of hot-plug linecards: cat \$bsp_path/config/hotplug_linecards		

Read Gearbox Counter

Node name	\$bsp_path/config/gearbox_counter		
Description	Get the number of gearboxes in the system.		
Access			
Release version	V.7.0010.3000		
Arguments	Name	Data type	Values
		Integer	0-X
Example	cat \$bsp_path/config/gearbox_counter		

Read Module Counter

Node name	\$bsp_path/config/module_counter		
Description	Get the number of sfp modules in the system Note: this is attribue is valid only for I2C ASIC driver		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer	1-X
Example	Get sfp module: cat \$bsp_path/config/module_counter		

Read ASIC Chipup Counter

Node name	\$bsp_path/config/asic_chipup_counter		
Description	Number of remaining ASIC chip-up retry attempts. This counter is decremented with each chip-up attempt and reset to the initial retry count when all attempts are exhausted.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
Example	Get asic chipup completed: <code>cat \$bsp_path/config/asic_chipup_counter</code>		

Read ASIC Chipup Completed

Node name	\$bsp_path/config/asic_chipup_completed		
Description	counter of successful ASIC driver initialization completions: 0 - no successful initialization completion. 1 - one ASIC device has been successful initialized. n - 'n' ASIC devices has been successful initialized.		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
Example	Status	Integer	1-X
Example	Get asic chipup completed: <code>cat \$bsp_path/config/asic_chipup_completed</code>		

Read Init Done

Node name	\$bsp_path/config/asics_init_done		
Description	is to be set to one, when 'asic_chipup_completed' attribute matches 'asic_num' attribute (old static attribute /var/run/hw-management/config/asic_num)		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer	1-X
Example	Get asics init done: cat \$bsp_path/config/asics_init_done		

Read Max System Fans (rotors)

Node name	\$bsp_path/config/max_tachos		
Description	Get max number of system fans.		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer	1-X
Example	Get fans max value: cat \$bsp_path/config/max_tachos		

Read Fan Drawer Number

Node name	\$bsp_path/config/fan_drwr_num		
Description	Get number of system FAN drawers		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer	1-X
Example	Get number of system FAN drawers: cat \$bsp_path/config/fan_drwr_num		

Read Fan Command

Node name	\$bsp_path/config/fan_command		
Description	Get PMBUS command for PSU config		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Hex	0xhh
Example	Get fan command: cat \$bsp_path/config/fan_command		

Read Fan Max/Min Speed

Node name	\$bsp_path/config/fan_max_speed \$bsp_path/config/fan_min_speed		
Description	Get the absolute system fan max/min speed		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer	X
Example	Get fan max speed: cat \$bsp_path/config/fan_max_speed Get fan min speed: cat \$bsp_path/config/fan_min_speed		

Read Fan Max/Min Speed for front/rear FAN

Node name	\$bsp_path/config/fan_front_max_speed \$bsp_path/config/fan_front_min_speed \$bsp_path/config/fan_rear_max_speed \$bsp_path/config/fan_rear_min_speed		
Description	Get the absolute system fan max/min speed for front/rear FAN. These attributes can be present on some switch types. If not present - fan_max_speed/fan_min_speed should be used instead.		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer	X
Example	Get front fan max speed: cat \$bsp_path/config/fan_front_max_speed Get front fan min speed: cat \$bsp_path/config/fan_front_min_speed		

Fan Speed Tolerance

Node name	\$bsp_path/config/fan_speed_tolerance		
Description	Max tolerance for measured FAN min/max speed compared to reference defined in fan_max_speed/fan_min_speed		
Access	Read only		
Release version	V.7.0040.1032		
Arguments	Name	Data type	Values
		int (percent)	0..90 (default 15)
Example	cat \$bsp_path/config/fan_speed_tolerance		

Fan Speed Units

Node name	\$bsp_path/config/fan_speed_units		
Description	Value to write PSU PMBUS register FAN_CONFIG_1_2 Set FAN Commanded in Duty Cycle. Used for PSU fan SET configuration		
Access	Read only		
Release version	V.7.10.3000		
Arguments	Name	Data type	Values
	Status	Integer	X (default 0x90)
Example	cat \$bsp_path/config/fan_speed_units i2cset -f -y "\$bus" "\$addr" "\$fan_config_command" "\$fan_speed_units" bp		

Number of Leakage Sensors

Node name	\$bsp_path/config/leakage_counter		
Description	Get the number of leakage sensors installed in the switch.		
Access	RO		
Release version	3.0		
Arguments	Name	Data type	Values
	Status	Integer	1-8
Example	<pre>cat \$bsp_path/config/leakage_counter</pre> Note: On SN6600_LD (SKU HI193), <code>sn66xxld_specific()</code> in <code>hw-management.sh</code> sets <code>leakage_count</code> to 2, which becomes <code>config/leakage_counter</code>. The same script initializes <code>events/leakage1</code> and <code>events/leakage2</code> only. Additional <code>system/leakage<N></code> symlinks (for example <code>leakage3</code> through <code>leakage5</code> in the validated tree) can appear when <code>mlxreg-io</code> exposes those <code>hwmon</code> attributes; <code>hw-management-thermal-events.sh</code> links each present <code>leakageN</code> up to its <code>max_leakage</code> limit (8). Consumers should treat <code>leakage_counter</code> as the supported event and policy count unless platform documentation states otherwise.		

Number of Cable Cartridges

Node name	\$bsp_path/config/cartridge_counter		
Description	Get the number of cable cartridges in the system. Cable cartridges are used in liquid-cooled multi-ASIC systems for connecting external cables. Note: This attribute is primarily for liquid-cooled systems with cable cartridges (N61XX_LD family: N6100_LD, N6300_LD). N6100_LD and N6300_LD (SKU H1185) have 4 cable cartridges (config/cartridge_counter = 4).		
Access	Read only		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	Count	Integer	0-X (number of cartridges)
Example	Get cartridge count: cat \$bsp_path/config/cartridge_counter		

Read/write Time Window for Thermal Control Periodic Log Report

Node name	\$bsp_path/config/periodic_report		
Description	Get/Set time for thermal control periodic log report (sec, default 7200)		
Access	Read/Write		
Release version	V.7.0010.1300		
Arguments	Name	Data type	Values
	Status	Integer	X
Example	Set periodic log report time: echo 3000 > \$bsp_path/config/periodic_report		

Read PSU I2C Address

Node name	\$bsp_path/config/psu<power supply module number>_i2c_addr		
Description	Get the I2C address of PSU for direct connection		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Hex	0xhh
Example	Get PSU1 I2C address: cat \$bsp_path/config/psu1_i2c_addr		

Read PSU I2C Bus

Node name	\$bsp_path/config/psu<X>_i2c_bus Where <X> is the power supply module index (for example psu1_i2c_bus, psu2_i2c_bus).		
Description	Contains the I2C bus number of PSU<X> used for direct connection to that power supply module. This configuration parameter is optional and is not mandatory on all platforms. It is created and used by the hw-management package for internal purposes (for example dummy PSU detection and mapping on supported switch systems). The OS and user-space applications must not assume that psu<X>_i2c_bus is always present; check for the file before reading it.		
Access	Read only Mandatory: No		
Release version	3.2		
Arguments	Name	Data type	Values
	Bus index	Integer	Platform-specific I2C bus number (for example 3, 4)
Example	Get PSU1 I2C bus when the file exists: if [-f \$bsp_path/config/psu1_i2c_bus]; then cat \$bsp_path/config/psu1_i2c_bus fi		

Read Thermal Delay

Node name	\$bsp_path/config/thermal_delay		
Description	Get the delay duration (seconds) since the HW mgmt service starts until thermal control init		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer (seconds)	X
Example	Get thermal delay: cat \$bsp_path/config/thermal_delay		

Dummy PSUs Supported

Node name	\$bsp_path/config/dummy_psus_supported		
Description	Indicates whether the system supports dummy PSUs for power management. Set to "1" for systems that support dummy PSU functionality, "0" otherwise.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
		Integer	0 or 1
Example	cat \$bsp_path/config/dummy_psus_supported 1		

Read PSU power capacity

Node name	\$bsp_path/config/psu[x]_power_capacity		
Description	Get the maximum power capacity for the psu.		
Access	Read only This attribute is present only in SN5600 and SN5400 platforms.		
Release version	1.0		
Arguments	Name	Data type	Values
	Constant	Milli watt	X
Example	Get power capacity of the psu cat \$bsp_path/config/psu[x]_power_capacity		

Read PSU power slope

Node name	\$bsp_path/config/psu[x]_power_slope		
Description	Get the power slope value for the psu. This is a hardware defined constant and will be used in power calculation		
Access	Read only This attribute is present only in SN5600 and SN5400 platforms.		
Release version	1.0		
Arguments	Name	Data type	Values
	Constant	Integer	X
Example	Get power slope for the psu cat \$bsp_path/config/psu[x]_power_slope		

Read DPU Number

Node name	\$bsp_path/config/dpu_num		
Description	Number of DPUs (1-4)		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer (number)	X
Example	Get dpu number: cat \$bsp_path/config/dpu_num		

Read DPU Board type

Node name	\$bsp_path/config/dpu_board_type		
Description	Whether DPU sensors are loaded static or dynamic		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	String	Static / dynamic
Example	Get dpu board type: cat \$bsp_path/config/dpu_board_type		

Read DPU board bus offset

Node name	\$bsp_path/config/dpu_brd_bus_offset		
Description	DPU i2c bus offset		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer (number)	X
Example	Get dpu i2c bus offset cat \$bsp_path/config/dpu_brd_bus_offset		

Read DPU bus offset

Node name	\$bsp_path/config/dpu_bus_off		
Description	I2c bus offset for the dpu		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	Integer (number)	X
Example	Get dpu bus offset number: cat \$bsp_path/config/dpu_bus_off		

Read DPU events

Node name	\$bsp_path/config/dpu_events		
Description	Events supported by DPUs		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	String	Ev1, ev2, etc.
Example	Get the events supported by DPU cat \$bsp_path/config/dpu_events		

Read DPU events to host

Node name	\$bsp_path/config/dpu_to_host_events		
Description	DPU events to host		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Status	String	X
Example	Get dpu events to host cat \$bsp_path/config/dpu_to_host_events		

Labels Ready

Node name	\$bsp_path/config/labels_ready		
Description	Label folder \$bsp_path/ui_tree ready to use		
Access	Read only		
Release version	V.7.0030.0958		
Arguments	Name	Data type	Values
			0 - labels init in progress 1 - labels init ready

Example

CPU Type

Node name	\$bsp_path/config/cpu_type		
Description	CPU type ID. Extracted from /proc/cpuinfo. Format: 0xXXYY XX - model num YY - cpu family		
Access	Read Only		
Release version			
Arguments	Name	Data type	Values
		Hex	
Example	cat \$bsp_path/config/cpu_type 0x656 cpu family : 6 model : 86 model name : Intel(R) Xeon(R) CPU D-1527 @ 2.20GHz		

Named Busses

Node name	\$bsp_path/config/named_busses		
Description	List of I2C bus idx/bus names separated by 'space'. Contains meaningful names for I2C busses on main board Present on some systems.		
Access	Read Only		
Release version	V.7.0020.3100		
Arguments	Name	Data type	Values
Example	asic1 2 pwr 4 vr1 5 amb1 7 vpd 8		

I2C Bus Offset

Node name	\$bsp_path/config/i2c_bus_offset		
Description	Base i2c bus idx. used for internal purposes		
Access			
Release version			
Arguments	Name	Data type	Values
Example	cat \$bsp_path/config/i2c_bus_offset 2		

I2C Bus Connect Devices

Node name	\$bsp_path/config/i2c_bus_connect_devices		
Description	List of i2c devices/names/bus/add used for internal purposes		
Access	Read only		
Release version			
Arguments	Name	Data type	Values
Example	xdpe12284 0x62 5 voltmon1 xdpe12284 0x64 5 voltmon2 ...		

I2C Bus Default Off EEPROM CPU

Node name	\$bsp_path/config/i2c_bus_def_off_eeprom_cpu		
Description	Offset of i2c bus for CPU VPD eeprom		
Access	Read only		
Release version			
Arguments	Name	Data type	Values
Example			

I2C Comex Mon Bus Default

Node name	\$bsp_path/config/i2c_comex_mon_bus_default		
Description	Default I2C bus number for COMEX monitoring devices. Used for voltage monitoring and other COMEX-related sensor readings.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
		Integer	Bus number
Example			

I2C SWB Bus

Node name	\$bsp_path/config/i2c_swb_bus		
Description	Absolute I2C adapter number for the switch-board (SWB) CPLD used for cartridge identity registers (rack id, topology id, tray id, slot id) at slave address 0x31. Platform-specific (for example HI176 CPU bus 18, HI180 CPU bus 53). Created from platform.json variables.i2c_swb_bus or from legacy *_specific() init. Used by hw-management-generate-dump.sh to produce archive member cpld_swb_cartridge_dump when the node exists (rack_id 13 bytes + ASCII, topology/tray/slot).		
Access	Read only		
Release version			
Arguments	Name	Data type	Values
		Integer	Bus number
Example	<pre>cat \$bsp_path/config/i2c_swb_bus 53 Example cpld_swb_cartridge_dump (from /tmp/hw-mgmt-dump.tar.gz): "text i2c_swb_bus=53 addr=0x31 rack_id: 0x31 0x38 0x32 0x34 0x32 0x32 0x35 0x34 0x31 0x30 0x30 0x31 0x33 rack_id_ascii: 1824225410013 topology_id: 0x00 tray_id: 0x00 slot_id: 0x01 "</pre>		

LM Sensors Configuration

Node name	\$bsp_path/config/lm_sensors_config		
Description	Configuration file for sensor tool from lm_sensors package. Contains sensors definition for the system		
Access			
Release version			
Arguments	Name	Data type	Values
		Text file	
Example	sensors -c \$bsp_path/config/lm_sensors_config		

LM Sensor Labels

Node name	\$bsp_path/config/lm_sensors_labels		
Description	Path to JSON file containing sensor labels for lm-sensors configuration. Provides human-readable names for various temperature, voltage, and fan sensors.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
		Text file	
Example	cat \$bsp_path/config/lm_sensors_labels /etc/hw-management-sensors/msn2700_sensors_labels.json		

Events Ready

Node name	\$bsp_path/config/events_ready		
Description	Provides indication that VPD parsing was completed.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
	Status	Integer	0/1
Example	Check whether VPD parsing has been completed <code>cat \$bsp_path/config/events_ready</code>		

Minimal Driver Unsupported

Node name	\$bsp_path/config/minimal_unsupported		
Description	Provides indication for whether ASIC I2C ('minimal') driver is supported.		
Access	Read Only		
Release version	3.0		
Arguments	Name	Data type	Values
	Status	Integer	0/1
Example	Get indication on whether minimal driver is supported: <code>\$bsp_path/config/minimal_unsupported</code>		

SGMII PHY

Node name	\$bsp_path/system/sgmii_phy		
Description	SGMII (Serial Gigabit Media Independent Interface) PHY status and configuration. Provides information about the physical layer interface for gigabit Ethernet connections.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
Example	cat \$bsp_path/system/sgmii_phy		

System Flow Capability

Node name	\$bsp_path/config/system_flow_capability		
Description	Indicates system flow capability for thermal management. Set to "C2P" (Cold to Power) for systems that support this flow direction.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
Example	cat \$bsp_path/config/system_flow_capability C2P		

Fan Direction EEPROM

Node name	\$bsp_path/config/fan_dir_eeprom		
Description	Enables fan direction detection from EEPROM. Set to "1" for systems that support fan direction detection via EEPROM. On these systems, each fan drawer direction (thermal/fan<index>_dir) is taken from the fan module EEPROM VPD. When the fan is removed, the corresponding fan<index>_dir attribute may also be removed. See Read Fan Direction.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
Example	cat \$bsp_path/config/fan_dir_eeprom 1		

Global Write Protection Wait Step

Node name	\$bsp_path/config/global_wp_wait_step		
Description	Write protection wait step configuration in seconds. Used by global write protection mechanism.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
Example	cat \$bsp_path/config/global_wp_wait_step 1		

Global Write Protection Timeout

Node name	\$bsp_path/config/global_wp_timeout		
Description	Write protection timeout configuration in seconds. Used by global write protection mechanism.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
Example	cat \$bsp_path/config/global_wp_timeout 20		

ConnectX Default I2C Bus

Node name	\$bsp_path/config/cx_default_i2c_bus		
Description	Default I2C bus number for ConnectX devices. Used for ConnectX device communication.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
Example	cat \$bsp_path/config/cx_default_i2c_bus 15		

JTAG Bridge Offset

Node name	\$bsp_path/config/jtag_bridge_offset		
Description	JTAG bridge memory offset for debugging purposes. Extracted from /proc/iomem mlxplat_jtag_bridge entry.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
Example	cat \$bsp_path/config/jtag_bridge_offset 0x40000000		

Core 0 Temperature ID

Node name	\$bsp_path/config/core0_temp_id		
Description	Temperature sensor ID for CPU core 0. Used for thermal monitoring of the first CPU core.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
Example	cat \$bsp_path/config/core0_temp_id 2		

Core 1 Temperature ID

Node name	\$bsp_path/config/core1_temp_id		
Description	Temperature sensor ID for CPU core 1. Used for thermal monitoring of the second CPU core.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
Example	cat \$bsp_path/config/core1_temp_id 3		

Read PDB Hotswap Scale Factor

Stack	Host		
Node name	\$bsp_path/config/pdb_hotswap_scale		
Description	LM5066I PDB hot-swap input power and current scaling factor. Written by sn66xxld_specific() in hw-management.sh for SN6600_LD (SKU HI193). The same value is symlinked under each lm5066i PDB hotswap environment node as *_power1_scale and *_curr1_scale (see §3.4). lm-sensors applies the same factor via compute rules in usr/etc/hw-management-sensors/sn66xxld_sensors.conf.		
Access	Read only		
Release version	V.7.0070.1000		
Arguments	Name	Data type	Values
	Scale	Float	5.333 on SN6600_LD (validated in hw-management.sh)
Example	cat \$bsp_path/config/pdb_hotswap_scale		

Read LED Control Type Map

Stack	Host		
Node name	\$bsp_path/config/led_control_type		
Description	Optional platform map of LED name (or glob mask) to control owner. Written by <code>set_config_data()</code> in <code>hw-management.sh</code> when a <code>*_specific()</code> function sets the <code>led_control_type</code> array. Space-separated pairs: <code>name type [name type ...]</code>. If the node is absent, or a given LED name is not listed, LED add uses the default owner <code>led_hw_sw</code> (see §3.16.28). Name matching (in <code>hw-management-chassis-events.sh get_led_control_type()</code>): 1. Exact match of the udev LED name (<code>status</code>, <code>fan</code>, <code>fan1</code>) or <code>led_<name></code>. <code>fan</code> and <code>fan1</code> are different names. 2. Glob mask: <code>*</code> matches any string, <code>?</code> matches one character. First matching mask wins. Masks must be quoted in the platform array (<code>"fan"</code>, <code>"led?"</code>) so the shell does not expand them. 3. Otherwise <code>led_hw_sw</code>.		
Access	Read only		
Release version	3.2.7		
Arguments	Name	Data type	Values
	name	String	LED name (<code>fan</code> , <code>status</code>) or mask (<code>fan*</code> , <code>led_psu?</code> , <code>led*</code>)
	type	String	<code>led_sw</code> , <code>led_hw</code> , <code>led_hw_sw</code>
Example	<pre>cat \$bsp_path/config/led_control_type # fan led_sw psu led_sw status led_sw</pre>		

Cable Cartridge FRU Validity

Stack	Host		
Node name	<code>\$bsp_path/config/cable_cartridge<cartridge number>_valid</code>		
Description	Result of the IPMI FRU check run on the cable cartridge EEPROM at init. The BMC reads the same EEPROM and programs the rack id, topology id, switch tray id and slot id into the switch board CPLD without validating it, so this node reports whether that source data can be trusted. Written by <code>validate_cartridge_fru()</code> in <code>hw-management-chassis-events.sh</code> after <code>ipmi-fru</code> parses <code>eeeprom/cable_cartridge<n>_eeeprom</code>. <code>ipmi-fru</code> exits 0 even for a broken FRU, so the verdict comes from its output: a FRU Error line, or a missing board serial number or chassis custom info field, gives 0. Failures are also reported to syslog. Created on the N51XX_LD and N61XX_LD families only. Other platforms that have cable cartridge EEPROMs parse them through a different path and do not get this node. It is also absent when <code>ipmi-fru</code> is not installed. The node is created when the cartridge EEPROM appears and removed when the cartridge is removed, so it never reports a verdict for an empty slot. One node per cartridge, so the count follows <code>config/cartridge_counter</code>, which is platform dependent: 4 on N6100_LD (HI180), N6300_LD (HI185), N5110_LD (HI166) and N5100_LD (HI167/HI170), and 2 on N5112_LD (HI169).		
Access	Read only		
Release version	3.2.8		
Arguments	Name	Data type	Values
	Validity	Integer	1 - FRU valid, 0 - FRU invalid
Example	Check cable cartridge 1 FRU validity: <code>cat \$bsp_path/config/cable_cartridge1_valid</code>		

BIOS Control

BIOS Active Image

Node name	\$bsp_path/system/bios_active_image		
Description	Currently active BIOS image identifier. Indicates which BIOS image is currently running on the system.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
		Text	Image identifier
Example	cat \$bsp_path/system/bios_active_image primary		

EEPROM Control

Read CPU EEPROM Data

Node name	\$bsp_path/eeprom/cpu_info		
Description	Read CPU raw data in hexadecimal format		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	EEPROM information	Hex	Hex dump format of memory
Example	Get CPU EEPROM information: cat \$bsp_path/eeprom/cpu_info		

Read Fan Module EEPROM Data

Node name	\$bsp_path/eeprom/fan<fan module number>_info		
Description	Read fan module raw data in hexadecimal format Note: This attribute is not supported on Comex CPU systems.		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	EEPROM information	Hex	Hex dump format of memory
Example	Get fan module 1 EEPROM information: <code>hexdump -C \$bsp_path/eeprom/fan1_info</code>		

Read Power Supply Module EEPROM Data

Node name	\$bsp_path/eeprom/psu<power supply module number>_info		
Description	Read power supply module raw data in hexadecimal format		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	EEPROM information	Hex	Hex dump format of memory
Example	Get power supply module 1 EEPROM information: <code>cat \$bsp_path/eeprom/psu1_info</code>		

Read System Chassis EEPROM Data

Node name	\$bsp_path/eeprom/vpd_info		
Description	Read system chassis raw data in hexadecimal format		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	EEPROM information	Hex	Hex dump format of memory
Example	Get system chassis EEPROM information: <code>cat \$bsp_path/eeprom/vpd_info</code>		

Read System Chassis EEPROM Parsed Data

Node name	\$bsp_path/eeprom/vpd_data		
Description	Read system chassis parsed data in text format		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	EEPROM information	string	string format of memory
Example	Get system chassis EEPROM information: <code>cat \$bsp_path/eeprom/vpd_data</code>		

Read Cable Cartridge EEPROM Data

Node name	<code>\$bsp_path/eeprom/cable_cartridge<index>_eeprom</code> <code>\$bsp_path/eeprom/cable_cartridge<index>_eeprom_data</code>		
Description	Read cable cartridge EEPROM raw data or parsed data. Cable cartridges are used in liquid-cooled multi-ASIC systems for connecting external cables. Note: This attribute is for systems with cable cartridges (N61XX_LD family: N6100_LD, N6300_LD). N6100_LD has 4 cable cartridges (index 1-4) on I2C buses 68, 69, 70, 71. N6300_LD (SKU H1185) uses the same cartridge EEPROM connect table as N6100_LD (<code>n61xxld_cartridge_eeprom_connect_table</code> in <code>hw-management.sh</code>).		
Access	Read only		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	index	Integer	1-4 (N6100_LD)
	EEPROM data	Binary/String	Raw or parsed EEPROM content
Example	<pre>Get cable cartridge EEPROM data: cat \$bsp_path/eeprom/cable_cartridge1_eeprom_data cat \$bsp_path/eeprom/cable_cartridge2_eeprom_data cat \$bsp_path/eeprom/cable_cartridge3_eeprom_data cat \$bsp_path/eeprom/cable_cartridge4_eeprom_data</pre>		

Read system EEPROM (BMC stack)

Stack	BMC (<code>hw-management-bmc-events.sh</code> , event <code>eeprom_system</code>)
Node name	<code>\$bsp_path/eeprom/eeprom_system</code>
Description	Raw system VPD EEPROM on the BMC image. H1189 udev matches I2C 5-0051 (<code>hw-management-bmc-early-i2c-devices.json</code>: bus 5 / 0x51, 24c512). Symlink points at the kernel <code>eeprom</code> attribute for that device.
Access	Read only
Release version	1.0
Example	<code>cat \$bsp_path/eeprom/eeprom_system</code>

Read BMC board EEPROM (BMC stack)

Stack	BMC (hw-management-bmc-events.sh, event eeprom_bmc)
Node name	\$bsp_path/eeprom/eeprom_bmc
Description	BMC FRU EEPROM on the BMC image. HI189 udev matches I2C 4-0050 (bus 4 / 0x50, 24c02 per early-I2C JSON).
Access	Read only
Release version	1.0
Example	cat \$bsp_path/eeprom/eeprom_bmc Reference: bmc/examples/hw-management-bmc-eeprom-sysfs.txt

Environment Control

Get A2D Voltage

Node name	\$bsp_path/environment/a2d_iio:device<number>_raw<index>		
Description	Get raw voltage input from A2D sensor		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Voltage	Integer	X
Example	Get voltage input from A2D1: cat \$bsp_path/environment/a2d_iio:device0_raw_1		

Get Comex Voltage Current

Node name	\$bsp_path/environment/comex_voltmon<index>_curr<index>_input		
Description	Get raw voltage input from Comex Note: This attribute is for Comex based system only		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Voltage	Integer	X
Example	Get comex voltage monitor 1 current2 reading: <code>cat \$bsp_path/environment/comex_voltmon1_curr2_input</code>		

Get Comex Voltage Input

Node name	\$bsp_path/environment/comex_voltmon<index>_in<index>_input		
Description	Get raw voltage input from Comex Note: This attribute is for Comex based system only		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Voltage	Integer	X
Example	Get Comex voltage monitor 1 input reading: <code>cat \$bsp_path/environment/comex_voltmon1_in1_input</code>		

Get Comex Voltage Power

Node name	\$bsp_path/environment/comex_voltmon<index>_power<index>_input		
Description	Get raw voltage input from Comex Note: This attribute is for Comex based system only		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Voltage	Integer	X
Example	Get Comex voltage monitor 1 power reading: <code>cat \$bsp_path/environment/comex_power2_input</code>		

Get System Voltage Current

Node name	\$bsp_path/environment/voltmon<index>_curr<index>_input		
Description	Get raw voltage input from system		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Voltage	Integer	X
Example	Get voltage monitor 1 current2 reading: <code>cat \$bsp_path/environment/voltmon1_curr2_input</code>		

Get System Voltage Input

Node name	\$bsp_path/environment/voltmon<index>_in<index>_input		
Description	Get raw voltage input from system		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Voltage	Integer	X
Example	Get voltage monitor 1 input reading: <code>cat \$bsp_path/environment/voltmon1_in1_input</code>		

Get System Voltage Power

Node name	\$bsp_path/environment/voltmon<index>_power<index>_input		
Description	Get raw voltage input from system		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	Voltage	Integer	X
Example	Get voltage monitor 1 power reading: <code>cat \$bsp_path/environment/voltmon1_power2_input</code>		

Get PDB Hotswap Controller Current

Node name	\$bsp_path/environment/pdb_hotswap<index>_curr<index>_input		
Description	Get PDB (Power Distribution Board) hot-swap controller current measurement Note: This attribute is for liquid-cooled systems only (SN58XX_LD family: SN5810_LD, SN5800_LD; N61XX_LD family: N6100_LD; SN66XX_LD family: SN6600_LD)		
Access	Read only		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Current	Integer	Value in milliamps (mA)
Example	Get PDB hotswap 1 current input: <code>cat \$bsp_path/environment/pdb_hotswap1_curr1_input</code>		

Get PDB Hotswap Controller Voltage

Node name	\$bsp_path/environment/pdb_hotswap<index>_in<index>_input		
Description	Get PDB hot-swap controller voltage measurement Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)		
Access	Read only		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Voltage	Integer	Value in millivolts (mV)
Example	Get PDB hotswap 1 input voltage: <code>cat \$bsp_path/environment/pdb_hotswap1_in1_input</code>		

Get PDB Hotswap Controller Power

Node name	\$bsp_path/environment/pdb_hotswap<index>_power<index>_input		
Description	Get PDB hot-swap controller power measurement Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)		
Access	Read only		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Power	Integer	Value in microwatts (µW)
Example	Get PDB hotswap 1 power input: <code>cat \$bsp_path/environment/pdb_hotswap1_power1_input</code> Note: On SN6600_LD, when the underlying hwmon device is lm5066i , <code>hw-management-chassis-events.sh</code> creates scale-factor symlinks for input power and current (see following sections). <code>lm-sensors</code> also applies factor 5.333 to the <code>lm5066i</code> PDB hotswap <code>power1</code> and <code>curr1</code> labels in <code>sn66xxld_sensors.conf</code> .		

Get PDB Hotswap Power Scale

Stack	Host
Node name	\$bsp_path/environment/pdb_hotswap<index>_power1_scale
Description	Scaling factor symlink for PDB hot-swap controller input power. Created when the hotswap hwmon driver name is lm5066i ; the symlink targets <code>\$bsp_path/config/pdb_hotswap_scale</code> . Note: SN6600_LD (SKU H1193) only. Not created for other hotswap driver types (for example mp5926).
Access	Read only
Release version	V.7.0070.1000
Example	<code>cat \$bsp_path/environment/pdb_hotswap1_power1_scale</code>

Get PDB Hotswap Current Scale

Stack	Host
Node name	<code>\$bsp_path/environment/pdb_hotswap<index>_curr1_scale</code>
Description	Scaling factor symlink for PDB hot-swap controller input current. Created when the hotswap hwmon driver name is Im5066i ; the symlink targets <code>\$bsp_path/config/pdb_hotswap_scale</code> . Note: SN6600_LD (SKU HI193) only.
Access	Read only
Release version	V.7.0070.1000
Example	<code>cat \$bsp_path/environment/pdb_hotswap1_curr1_scale</code>

Get PDB Hotswap Controller Thresholds

Node name	<code>\$bsp_path/environment/pdb_hotswap<index>_<sensor>_<threshold></code>		
Description	Get PDB hot-swap controller threshold values (crit, lcrit, max, min) Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)		
Access	Read only		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Threshold	Integer	Varies by sensor type
Example	Get PDB hotswap 1 current max threshold: <code>cat \$bsp_path/environment/pdb_hotswap1_curr1_max</code> <code>cat \$bsp_path/environment/pdb_hotswap1_in1_crit</code> <code>cat \$bsp_path/environment/pdb_hotswap1_in1_lcrit</code>		

Get PDB Power Converter Current

Node name	<code>\$bsp_path/environment/pdb_pwr_conv<index>_curr<index>_input</code> (SN58XX_LD, SN66XX_LD) <code>\$bsp_path/environment/pwr_conv<index>_curr<index>_input</code> (N61XX_LD)		
Description	Get PDB power converter current measurement (input or output) Note: This attribute is for liquid-cooled systems only. - SN58XX_LD family uses: <code>pdb_pwr_conv<N>_...</code> naming (1 power converter on SN5810_LD; more on SN5800_LD) - SN66XX_LD family (SN6600_LD) uses: <code>pdb_pwr_conv<N>_...</code> with N = 1..2 - N61XX_LD family uses: <code>pwr_conv<N>_...</code> naming (2 power converters)		
Access	Read only		
Release version	V.7.0050.3000 (SN58XX_LD), V.7.0060.1000 (N61XX_LD)		
Arguments	Name	Data type	Values
	Current	Integer	Value in milliamps (mA)
Example	Get power converter current readings: # SN58XX_LD systems <code>cat \$bsp_path/environment/pdb_pwr_conv1_curr1_input cat \$bsp_path/environment/pdb_pwr_conv1_curr2_input</code> # SN6600_LD systems (2 PDB converters) <code>cat \$bsp_path/environment/pdb_pwr_conv1_curr1_input cat \$bsp_path/environment/pdb_pwr_conv1_curr2_input cat \$bsp_path/environment/pdb_pwr_conv2_curr1_input cat \$bsp_path/environment/pdb_pwr_conv2_curr2_input</code> # N6100_LD systems (2 power converters) <code>cat \$bsp_path/environment/pwr_conv1_curr1_input cat \$bsp_path/environment/pwr_conv1_curr2_input cat \$bsp_path/environment/pwr_conv2_curr1_input cat \$bsp_path/environment/pwr_conv2_curr2_input</code>		

Get PDB Power Converter Voltage

Node name	<code>\$bsp_path/environment/pdb_pwr_conv<index>_in<index>_input</code> (SN58XX_LD, SN66XX_LD) <code>\$bsp_path/environment/pwr_conv<index>_in<index>_input</code> (N61XX_LD)		
Description	Get PDB power converter voltage measurement (input or output) Note: This attribute is for liquid-cooled systems only. - SN58XX_LD and SN66XX_LD families use: <code>pdb_pwr_conv<N>_...</code> naming - N61XX_LD family uses: <code>pwr_conv<N>_...</code> naming		
Access	Read only		
Release version	V.7.0050.3000 (SN58XX_LD), V.7.0060.1000 (N61XX_LD)		
Arguments	Name	Data type	Values
	Voltage	Integer	Value in millivolts (mV)
Example	Get power converter voltage readings: # SN58XX_LD systems <code>cat \$bsp_path/environment/pdb_pwr_conv1_in1_input cat \$bsp_path/environment/pdb_pwr_conv1_in2_input</code> # SN6600_LD systems <code>cat \$bsp_path/environment/pdb_pwr_conv1_in1_input cat \$bsp_path/environment/pdb_pwr_conv1_in2_input cat \$bsp_path/environment/pdb_pwr_conv2_in1_input cat \$bsp_path/environment/pdb_pwr_conv2_in2_input</code> # N6100_LD systems <code>cat \$bsp_path/environment/pwr_conv1_in1_input cat \$bsp_path/environment/pwr_conv1_in2_input cat \$bsp_path/environment/pwr_conv2_in1_input cat \$bsp_path/environment/pwr_conv2_in2_input</code>		

Get PDB Power Converter Power

Node name	<code>\$bsp_path/environment/pdb_pwr_conv<index>_power<index>_input</code> (SN58XX_LD, SN66XX_LD) <code>\$bsp_path/environment/pwr_conv<index>_power<index>_input</code> (N61XX_LD)		
Description	Get PDB power converter power measurement (input or output) Note: This attribute is for liquid-cooled systems only. - SN58XX_LD and SN66XX_LD families use: <code>pdb_pwr_conv<N>_...</code> naming - N61XX_LD family uses: <code>pwr_conv<N>_...</code> naming		
Access	Read only		
Release version	V.7.0050.3000 (SN58XX_LD), V.7.0060.1000 (N61XX_LD)		
Arguments	Name	Data type	Values
	Power	Integer	Value in microwatts (µW)
Example	Get power converter power readings: # SN58XX_LD systems <code>cat \$bsp_path/environment/pdb_pwr_conv1_power1_input</code> # SN6600_LD systems <code>cat \$bsp_path/environment/pdb_pwr_conv1_power1_input</code> <code>cat \$bsp_path/environment/pdb_pwr_conv2_power1_input</code> # N6100_LD systems <code>cat \$bsp_path/environment/pwr_conv1_power1_input</code> <code>cat \$bsp_path/environment/pwr_conv2_power1_input</code>		

Get PDB Power Converter Thresholds

Node name	\$bsp_path/environment/pdb_pwr_conv<index>_<sensor>_<threshold>		
Description	Get PDB power converter threshold values (crit, lcrit, max, min) Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)		
Access	Read only		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Threshold	Integer	Varies by sensor type
Example	Get PDB power converter 1 thresholds: cat \$bsp_path/environment/pdb_pwr_conv1_curr1_crit cat \$bsp_path/environment/pdb_pwr_conv1_in1_max cat \$bsp_path/environment/pdb_pwr_conv1_in1_min		

Events

Get FAN hot-plug event status

Node name	\$bsp_path/events/fan<index>		
Description	Get hot-plug event status of FAN<index> Index max value corresponds to \$bsp_path/config/hotplug_fans 0 – FAN<index> was removed, 1 – FAN<index> was inserted.		
Access	Read		
Release version	V.7.0010.1300		
Arguments	Name	Data type	Values
	Thermal	Integer	0 / 1
Example	Get FAN3 hot-plug status: cat \$bsp_path/events/fan3		

Get PSU hot-plug event status

Node name	\$bsp_path/events/psu<index>		
Description	Get hot-plug event status of PSU<index> Index max value corresponds to \$bsp_path/config/hotplug_psus 0 – PSU<index> was removed, 1 – PSU<index> was inserted.		
Access	Read		
Release version	V.7.0010.1300		
Arguments	Name	Data type	Values
	Thermal	Integer	0 / 1
Example	Get PSU2 hot-plug status: cat \$bsp_path/events/psu2		

PWR hot-plug event status

Node name	\$bsp_path/events/pwr<index>		
Description	Get latest hot-plug event status of PWR<index> Index max value corresponds to \$bsp_path/config/hotplug_pwrs 0 – PWR<index> cable was plugged-out, 1 – PWR<index> cable was plugged-in.		
Access	Read		
Release version	V.7.0010.1300		
Arguments	Name	Data type	Values
	Thermal	Integer	0 / 1
Example	Get Power1 cable hot-plug status: cat \$bsp_path/events/pwr1		

PDB hot-plug event status

Node name	\$bsp_path/events/pdb<index>		
Description	Get hot-plug event status of PDB (Power Distribution Board) <index> Index max value corresponds to \$bsp_path/config/hotplug_pdb 0 – PDB<index> was removed, 1 – PDB<index> was inserted. Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)		
Access	Read		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Event	Integer	0 / 1
Example	Get PDB1 hot-plug status: cat \$bsp_path/events/pdb1		

DPU Ready event

Node name	\$bsp_path/events/dpu[1-8]_ready		
Description	Get dpu ready event		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	0 / 1
Example	Get dpu[1-8] ready status cat \$bsp_path/events/dpu1_ready		

DPU Shutdown event

Node name	\$bsp_path/events/dpu[1-8]_shtdn_ready		
Description	Get dpu shutdown ready event		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	0 / 1
Example	Get dpu[1-8] shut down ready status cat \$bsp_path/events/dpu1_shtdn_ready		

EROT (External Root of Trust) AP event

Node name	\$bsp_path/events/erot<index>_ap		
Description	EROT Application Processor event status. Indicates the state of the External Root of Trust application processor at the specified index. Note: This attribute is for systems with eRoT support (N61XX_LD family: N6100_LD). N6100_LD has 1 eRoT (index=1).		
Access	Read only		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	index	Integer	1 (N6100_LD)
	Status	Integer	0 (inactive), 1 (active)
Example	Get eRoT AP status: cat \$bsp_path/events/erot1_ap		

EROT (External Root of Trust) Error Event

Node name	\$bsp_path/events/erot<index>_error		
Description	EROT error event status. Indicates error conditions detected by the External Root of Trust at the specified index. Note: This attribute is for systems with eRoT support (N61XX_LD family: N6100_LD). N6100_LD has 1 eRoT (index=1).		
Access	Read only		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	index	Integer	1 (N6100_LD)
	Status	Integer	0 (no error), 1 (error detected)
Example	Get eRoT error status: cat \$bsp_path/events/erot1_error		

Graceful Power Off Event

Node name	\$bsp_path/events/graceful_pwr_off		
Description	Graceful power off event status. Indicates when the system is performing a graceful shutdown sequence.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
		Integer	0 (normal), 1 (graceful shutdown)
Example			

Power Button Event

Node name	\$bsp_path/events/power_button		
Description	Power button event status. Indicates when the system power button has been pressed.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
		Integer	0 (not pressed), 1 (pressed)
Example			

Power Events

Node name	\$bsp_path/config/power_events		
Description	Power events configuration file. Contains settings and thresholds for power-related event handling.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
		Text file	Configuration data
Example			

Power Button Event

Node name	\$bsp_path/system/power_button_evt		
Description	Power button event status. Indicates the current state of power button events in the system.		
Access	Read only		
Release version	3.0		
Arguments	Name	Data type	Values
		Integer	Event status
Example	cat \$bsp_path/system/power_button_evt		

Alarms

Get PMBUS voltmon alarm status.

Node name	\$bsp_path/alarm/voltmon<index>_<sensor_name>_alarm		
Description	Get voltmonitor alarm status of voltmon<index> sensor_name one of: in, curr, power, temp. sensor count can be different for different voltmonitor types/configuration. Alarm set by voltmon sensor itself (hw controlled attribute) 1 – alarm set 0 – alarm clear		
Access	Read		
Release version	V.7.0010.1000		
Arguments	Name	Data type	Values
	Thermal	Integer	0 / 1
Example	Get voltmon1_in1_alarm: cat \$bsp_path/alarm/voltmon1_in1_alarm		

Get COMEX PMBUS voltmon alarm status.

Node name	\$bsp_path/alarm/comex_voltmon<index>_<sensor_name>_alarm		
Description	Get voltmonitor alarm status of comex_voltmon<index> sensor_name one of: in, curr, power, temp. sensor count can be different for different voltmonitor types/configuration. Alarm set by voltmon sensor itself (hw controlled attribute) 1 – alarm set 0 – alarm clear		
Access	Read		
Release version	V.7.0010.1000		
Arguments	Name	Data type	Values
	Thermal	Integer	0 / 1
Example	Get comex_voltmon1_in1_alarm: cat \$bsp_path/alarm/comex_voltmon1_in1_alarm		

Get PSU PMBUS alarm status.

Node name	\$bsp_path/alarm/psu<index>_<sensor_name>_alarm		
Description	Get PSU PMBUS alarm status of PSU<index> sensor_name: in, curr, power, temp. sensor count can be different for different PSU types/configuration. Alarm set by PSU sensor itself (hw controlled attribute) 1 – alarm set 0 – alarm clear		
Access	Read		
Release version	V.7.0010.1000		
Arguments	Name	Data type	Values
	Thermal	Integer	0 / 1
Example	Get psu1_power1_alarm: cat \$bsp_path/alarm/psu1_power1_alarm		

Get PDB Hotswap Controller alarm status

Node name	\$bsp_path/alarm/pdb_hotswap<index>_<sensor_name>_alarm		
Description	Get PDB (Power Distribution Board) hot-swap controller alarm status sensor_name: in, curr, power, temp. Alarm set by PDB hotswap controller sensor itself (hardware-controlled attribute) 1 – alarm set 0 – alarm clear Note: This attribute is for liquid-cooled systems only (SN58XX_LD family: SN5810_LD, SN5800_LD; N61XX_LD family: N6100_LD; SN66XX_LD family: SN6600_LD)		
Access	Read		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Alarm	Integer	0 / 1
Example	Get PDB hotswap 1 alarms: cat \$bsp_path/alarm/pdb_hotswap1_curr1_alarm cat \$bsp_path/alarm/pdb_hotswap1_in1_alarm cat \$bsp_path/alarm/pdb_hotswap1_power1_alarm cat \$bsp_path/alarm/pdb_hotswap1_temp1_crit_alarm cat \$bsp_path/alarm/pdb_hotswap1_temp1_max_alarm		

Get PDB Power Converter alarm status

Node name	\$bsp_path/alarm/pdb_pwr_conv<index>_<sensor_name>_alarm		
Description	Get PDB power converter alarm status sensor_name: in, curr, power, temp. Alarm set by PDB power converter sensor itself (hardware-controlled attribute) 1 – alarm set 0 – alarm clear Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)		
Access	Read		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Alarm	Integer	0 / 1
Example	Get PDB power converter 1 alarms: cat \$bsp_path/alarm/pdb_pwr_conv1_curr1_alarm cat \$bsp_path/alarm/pdb_pwr_conv1_curr2_alarm cat \$bsp_path/alarm/pdb_pwr_conv1_in1_alarm cat \$bsp_path/alarm/pdb_pwr_conv1_in2_alarm cat \$bsp_path/alarm/pdb_pwr_conv1_power1_alarm cat \$bsp_path/alarm/pdb_pwr_conv1_temp1_crit_alarm cat \$bsp_path/alarm/pdb_pwr_conv1_temp1_max_alarm		

Get CPU temp alarm status.

Node name	\$bsp_path/alarm/cpu_core<idx>_crit_alarm		
Description	Get CPU core overtemperature alarm status. idx – 0...cpu_core_num cpu_core_num can be different for different CPU type Alarm set by CPU sensor itself (hw controlled attribute) 1 – alarm set 0 – alarm clear		
Access	Read		
Release version	V.7.0010.1000		
Arguments	Name	Data type	Values
	Thermal	Integer	0 / 1
Example	Get cpu_core0_crit_alarm: cat \$bsp_path/alarm/cpu_core0_crit_alarm		

PSU FW

Get Secondary FW version of PSU

Node name	\$bsp_path/firmware/psu<index>_fw_ver		
Description	Get secondary FW version of PSU<index> For Murata 1500/2000 and Delta 550 the contents of the file is the relevant FW version For all other PSUs - the contents is string "N/A"		
Access	Read		
Release version	V.7.0020.2000		
Arguments	Name	Data type	Values
	version	string	
Example	Get secondary FW version of PSU1 \$bsp_path/firmware/psu1_fw_ver		

Get Primary FW version of PSU

Node name	\$bsp_path/firmware/psu<index>_fw_primary_ver		
Description	Get primary FW version of PSU<index> Primary files exist only for Murata. For all other PSUs - the contents is string "N/A"		
Access	Read		
Release version	V.7.0020.2000		
Arguments	Name	Data type	Values
	version	string	
Example	Get primary FW version of PSU1 \$bsp_path/firmware/psu1_fw_primary_ver		

DPU system attributes

Get DPU id

Node name	\$bsp_path/system/dpu<index>_id		
Description	Get DPU<index> id		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 id \$bsp_path/system/dpu1_id		

Get DPU boot progress

Node name	\$bsp_path/system/dpu<index>_boot_progress		
Description	Get DPU<index> boot progress		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 boot progress \$bsp_path/system/dpu1_boot_progress		

Get DPU cpld version

Node name	\$bsp_path/system/dpu<index>_cpld_version		
Description	Get DPU<index> CPLD version		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 cpld version \$bsp_path/system/dpu1_cpld_version		

Get DPU cpld base version

Node name	\$bsp_path/system/dpu<index>_cpld_base_version		
Description	Get DPU<index> CPLD base version		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 cpld base version \$bsp_path/system/dpu1_cpld_base_version		

Get DPU auxiliary power reset reason

Node name	\$bsp_path/system/dpu<index>_aux_pwr_rst_reason		
Description	Get DPU<index> auxiliary power reset reason		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 auxiliary power reset reason \$bsp_path/system/dpu1_aux_pwr_rst_reason		

Get DPU main board reset reason

Node name	\$bsp_path/system/dpu<index>_main_brd_rst_reason		
Description	Get DPU<index> main board reset reason		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 main board reset reason \$bsp_path/system/dpu1_main_brd_rst_reason		

Get DPU comex power failure

Node name	\$bsp_path/system/dpu<index>_comex_pwr_fail		
Description	Get DPU<index> comex power failure		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 comex power failure \$bsp_path/system/dpu1_comex_pwr_fail		

Get DPU power reset reason

Node name	\$bsp_path/system/dpu<index>_pwr_rst_reason		
Description	Get DPU<index> power reset reason		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 power reset reason \$bsp_path/system/dpu1_pwr_rst_reason		

Get DPU thermal shutdown reason

Node name	\$bsp_path/system/dpu<index>_thermal_shtdn_reason		
Description	Get DPU<index> thermal shutdown reason		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 thermal shutdown reason \$bsp_path/system/dpu1_thermal_shtdn_reason		

Get DPU tpm reset reason

Node name	\$bsp_path/system/dpu<index>_tpm_rst_reason		
Description	Get DPU<index> tpm reset reason		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 tpm reset reason \$bsp_path/system/dpu1_tpm_rst_reason		

Get DPU perst reset reason

Node name	\$bsp_path/system/dpu<index>_perst_rst_reason		
Description	Get DPU<index> perst reset reason		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 perst reset reason \$bsp_path/system/dpu1_perst_rst_reason		

Get DPU phy reset reason

Node name	\$bsp_path/system/dpu<index>_phy_rst_reason		
Description	Get DPU<index> phy reset reason		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 phy reset reason \$bsp_path/system/dpu1_phy_rst_reason		

Get DPU usb phy reset reason

Node name	\$bsp_path/system/dpu<index>_usb_phy_rst_reason		
Description	Get DPU<index> usb phy reset reason		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 usb phy reset reason \$bsp_path/system/dpu1_usb_phy_rst_reason		

Get DPU fpga part number

Node name	\$bsp_path/system/dpu<index>_fpga_part_number		
Description	Get DPU<index> fpga part number		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 fpga part number \$bsp_path/system/dpu1_fpga_part_number		

Get DPU fpga minor version

Node name	\$bsp_path/system/dpu<index>_fpga_minor_version		
Description	Get DPU<index> fpga minor version		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 fpga minor version \$bsp_path/system/dpu1_fpga_minor_version		

Get DPU ufm upgrade status

Node name	\$bsp_path/system/dpu<index>_ufm_upgrade_status		
Description	Get DPU<index> ufm upgrade status		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 ufm upgrade status \$bsp_path/system/dpu1_ufm_upgrade_status		

Get DPU VR update status

Node name	\$bsp_path/system/dpu<index>_vr_update_status		
Description	Get DPU<index> VR update status		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 VR update status \$bsp_path/system/dpu1_vr_update_status		

DPU events

Get DPU PLL power good indication

Node name	\$bsp_path/events/dpu<index>_pll_pwr_good		
Description	Get DPU<index> PLL power good indication		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 PLL power good indication \$bsp_path/events/dpu1_pll_pwr_good		

Get DPU input power indication

Node name	\$bsp_path/events/dpu<index>_input_pwr		
Description	Get DPU<index> input power indication		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 input power indication \$bsp_path/events/dpu1_input_pwr		

Get DPU serdes power indication

Node name	\$bsp_path/events/dpu<index>_serdes_pwr		
Description	Get DPU<index> serdes power indication		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 serdes power indication \$bsp_path/events/dpu1_serdes_pwr		

Get DPU serdes analog power indication

Node name	\$bsp_path/events/dpu<index>_serdes_analog_pwr		
Description	Get DPU<index> serdes analog power indication		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 serdes analog power indication \$bsp_path/events/dpu1_serdes_analog_pwr		

Get DPU core power indication

Node name	\$bsp_path/events/dpu<index>_core_pwr		
Description	Get DPU<index> core power indication		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 core power indication \$bsp_path/events/dpu1_core_pwr		

Get DPU cpu power indication

Node name	\$bsp_path/events/dpu<index>_cpu_pwr		
Description	Get DPU<index> cpu power indication		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 cpu power indication \$bsp_path/events/dpu1_cpu_pwr		

Get DPU digital interfaces power

Node name	\$bsp_path/events/dpu<index>_digital_if_pwr		
Description	Get DPU<index> digital interfaces power		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 digital interfaces power \$bsp_path/events/dpu1_digital_if_pwr		

Get DPU ddr5 power indication

Node name	\$bsp_path/events/dpu<index>_ddr5_pwr		
Description	Get DPU<index> ddr5 power indication		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 ddr5 power indication \$bsp_path/events/dpu1_ddr5_pwr		

Get DPU thermal trip indication

Node name	\$bsp_path/events/dpu<index>_thermal_trip		
Description	Get DPU<index> thermal trip indication		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 thermal trip indication \$bsp_path/events/dpu1_thermal_trip		

Get DPU tps upgrade status

Node name	\$bsp_path/events/dpu<index>_tps_upgrade		
Description	Get DPU<index> tps upgrade status		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 tps upgrade status \$bsp_path/events/dpu1_tps_upgrade		

Get DPU cpu power fault indication

Node name	\$bsp_path/events/dpu<index>_cpu_pwr_fault		
Description	Get DPU<index> cpu power fault indication		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 cpu power fault indication \$bsp_path/events/dpu1_cpu_pwr_fault		

Get DPU cpu VR hot alert

Node name	\$bsp_path/events/dpu<index>_cpu_vr_hot		
Description	Get DPU<index> cpu VR hot alert		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 cpu VR hot alert \$bsp_path/events/dpu1_cpu_vr_hot		

Get DPU ddr5 fault indication

Node name	\$bsp_path/events/dpu<index>_ddr5_fault		
Description	Get DPU<index> ddr5 fault indication		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 ddr5 fault indication \$bsp_path/events/dpu1_ddr5_fault		

Get DPU ddr5 hot alert

Node name	\$bsp_path/events/dpu<index>_ddr5_hot		
Description	Get DPU<index> ddr5 hot alert		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	dpu	Integer	
Example	Get dpu1 ddr5 hot alert \$bsp_path/events/dpu1_ddr5_hot		

LC Alarms

Get LC Hot Swap Power Alarm

Node name	\$bsp_path/alarm/lc<index>_hotswap_pwr_alarm		
Description	Get LC<index> hot swap power alarm		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get lc1 hot swap power alarm \$bsp_path/alarm/lc1_hotswap_pwr_alarm		

Get LC Voltage Input Alarm

Node name	\$bsp_path/alarm/lc<index>_volt_in_alarm		
Description	Get LC<index> voltage input alarm		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get lc1 voltage input alarm \$bsp_path/alarm/lc1_volt_in_alarm		

Get LC Voltage Current Alarm

Node name	\$bsp_path/alarm/lc<index>_volt_curr_alarm		
Description	Get LC<index> voltage current alarm		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get lc1 voltage current alarm \$bsp_path/alarm/lc1_volt_curr_alarm		

Get LC Voltage Power Alarm

Node name	\$bsp_path/alarm/lc<index>_volt_pwr_alarm		
Description	Get LC<index> voltage power alarm		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get lc1 voltage power alarm \$bsp_path/alarm/lc1_volt_pwr_alarm		

LC EEPROM

Read LC EEPROM FRU

Node name	\$bsp_path/eeprom/lc<index>_fru		
Description	Read LC<index> EEPROM FRU data in hexadecimal format		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	EEPROM information	Hex	Hex dump format of memory
Example	Get LC1 EEPROM FRU information: cat \$bsp_path/eeprom/lc1_fru		

Read LC EEPROM INI

Node name	\$bsp_path/eeprom/lc<index>_ini		
Description	Read LC<index> EEPROM INI data in hexadecimal format		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	EEPROM information	Hex	Hex dump format of memory
Example	Get LC1 EEPROM INI information: cat \$bsp_path/eeprom/lc1_ini		

Read LC EEPROM VPD Parsed

Node name	\$bsp_path/eeprom/lc<index>_vpd_parsed		
Description	Read LC<index> EEPROM VPD parsed data in text format		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	EEPROM information	string	Parsed VPD data
Example	Get LC1 EEPROM VPD parsed information: cat \$bsp_path/eeprom/lc1_vpd_parsed		

Read LC EEPROM INI Parsed

Node name	\$bsp_path/eeprom/lc<index>_ini_parsed		
Description	Read LC<index> EEPROM INI parsed data in text format		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	EEPROM information	string	Parsed INI data
Example	Get LC1 EEPROM INI parsed information: <code>cat \$bsp_path/eeprom/lc1_ini_parsed</code>		

LC Environment

Get LC Voltage Current

Node name	\$bsp_path/environment/lc<index>_volt_curr<idx>_input		
Description	Get LC<index> voltage current reading		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 voltage current reading: <code>cat \$bsp_path/environment/lc1_volt_curr1_input</code>		

Get LC Voltage Input

Node name	\$bsp_path/environment/lc<index>_volt_in<idx>_input		
Description	Get LC<index> voltage input reading		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 voltage input reading: cat \$bsp_path/environment/lc1_volt_in1_input		

Get LC Voltage Power

Node name	\$bsp_path/environment/lc<index>_volt_pwr<idx>_input		
Description	Get LC<index> voltage power reading		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 voltage power reading: cat \$bsp_path/environment/lc1_volt_pwr1_input		

Get LC Hot Swap Current

Node name	\$bsp_path/environment/lc<index>_hotswap_curr<idx>_input		
Description	Get LC<index> hot swap current reading		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 hot swap current reading: <code>cat \$bsp_path/environment/lc1_hotswap_curr1_input</code>		

Get LC Hot Swap Input

Node name	\$bsp_path/environment/lc<index>_hotswap_in<idx>_input		
Description	Get LC<index> hot swap input reading		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 hot swap input reading: <code>cat \$bsp_path/environment/lc1_hotswap_in1_input</code>		

Get LC Hot Swap Power

Node name	\$bsp_path/environment/lc<index>_hotswap_pwr<idx>_input		
Description	Get LC<index> hot swap power reading		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 hot swap power reading: <code>cat \$bsp_path/environment/lc1_hotswap_pwr1_input</code>		

Get LC A2D Voltage

Node name	\$bsp_path/environment/lc<index>_a2d_volt<idx>_input		
Description	Get LC<index> A2D voltage reading		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 A2D voltage reading: <code>cat \$bsp_path/environment/lc1_a2d_volt1_input</code>		

Get LC A2D Voltage Scale

Node name	\$bsp_path/environment/lc<index>_a2d_volt<idx>_scale		
Description	Get LC<index> A2D voltage scale		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 A2D voltage scale: cat \$bsp_path/environment/lc1_a2d_volt1_scale		

LC LED

Common LED API reference: LED_Control_API.md — line card paths under \$bsp_path/lc<N>/led/.

Get LC Status LED

Node name	\$bsp_path/led/lc<index>_status_led		
Description	Get LC<index> status LED state		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 status LED: cat \$bsp_path/led/lc1_status_led		

Get LC Status LED Capabilities

Node name	\$bsp_path/led/lc<index>_status_led_capability		
Description	Get LC<index> status LED capabilities		
Access	Read		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 status LED capabilities: cat \$bsp_path/led/lc1_status_led_capability		

Set LC Status Green/Orange

Node name	\$bsp_path/led/lc<index>_status_led		
Description	Set LC<index> status LED to green or orange		
Access	Write		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
	color	String	green, orange
Example	Set LC1 status LED to green: echo green > \$bsp_path/led/lc1_status_led		

Set LC Status LED Green/Orange Delay Off

Node name	\$bsp_path/led/lc<index>_status_led_delay_off		
Description	Set LC<index> status LED delay off time		
Access	Write		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
	delay	Integer	milliseconds
Example	Set LC1 status LED delay off to 1000ms: echo 1000 > \$bsp_path/led/lc1_status_led_delay_off		

Set LC Status LED Green/Orange Delay On

Node name	\$bsp_path/led/lc<index>_status_led_delay_on		
Description	Set LC<index> status LED delay on time		
Access	Write		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
	delay	Integer	milliseconds
Example	Set LC1 status LED delay on to 500ms: echo 500 > \$bsp_path/led/lc1_status_led_delay_on		

LC Config

Read LC CPLD Number

Node name	\$bsp_path/config/lc<index>_cpld_num		
Description	Get LC<index> CPLD number		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 CPLD number: cat \$bsp_path/config/lc1_cpld_num		

Read LC FPGA Number

Node name	\$bsp_path/config/lc<index>_fpga_num		
Description	Get LC<index> FPGA number		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 FPGA number: cat \$bsp_path/config/lc1_fpga_num		

Read LC Gearbox Number

Node name	\$bsp_path/config/lc<index>_gearbox_num		
Description	Get LC<index> gearbox number		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 gearbox number: cat \$bsp_path/config/lc1_gearbox_num		

Read LC Gearbox Manager Number

Node name	\$bsp_path/config/lc<index>_gearbox_mgr_num		
Description	Get LC<index> gearbox manager number		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 gearbox manager number: cat \$bsp_path/config/lc1_gearbox_mgr_num		

Read LC Port Number

Node name	\$bsp_path/config/lc<index>_port_num		
Description	Get LC<index> port number		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 port number: cat \$bsp_path/config/lc1_port_num		

Read LC Module Counter

Node name	\$bsp_path/config/lc<index>_module_counter		
Description	Get LC<index> module counter		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 module counter: cat \$bsp_path/config/lc1_module_counter		

UART

Node name	\$bsp_path/system/lc<index>_uart		
Description	Get/set LC<index> UART configuration		
Access	Read/Write		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 UART configuration: cat \$bsp_path/system/lc1_uart		

ASIC control

Node name	\$bsp_path/system/lc<index>_asic_control		
Description	Get/set LC<index> ASIC control		
Access	Read/Write		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 ASIC control: cat \$bsp_path/system/lc1_asic_control		

LC thermal

Read LC Gearbox Temperature Input

Node name	\$bsp_path/thermal/lc<index>_gearbox<idx>_temp_input		
Description	Get LC<index> gearbox<idx> temperature input		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 gearbox1 temperature: cat \$bsp_path/thermal/lc1_gearbox1_temp_input		

Get LC QSFP/SFP Module Thermal

Node name	\$bsp_path/thermal/lc<index>_module<idx>_temp_input		
Description	Get LC<index> QSFP/SFP module<idx> temperature		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 module1 temperature: cat \$bsp_path/thermal/lc1_module1_temp_input		

Read Temperature Critical Module

Node name	\$bsp_path/thermal/lc<index>_module<idx>_temp_crit		
Description	Get LC<index> module<idx> critical temperature threshold		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 module1 critical temperature: cat \$bsp_path/thermal/lc1_module1_temp_crit		

Read Temperature Emergency Module

Node name	\$bsp_path/thermal/lc<index>_module<idx>_temp_emergency		
Description	Get LC<index> module<idx> emergency temperature threshold		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 module1 emergency temperature: cat \$bsp_path/thermal/lc1_module1_temp_emergency		

Read Temperature Fault Module

Node name	\$bsp_path/thermal/lc<index>_module<idx>_temp_fault		
Description	Get LC<index> module<idx> temperature fault status		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 module1 temperature fault: <code>cat \$bsp_path/thermal/lc1_module1_temp_fault</code>		

Read Temperature Input Module

Node name	\$bsp_path/thermal/lc<index>_module<idx>_temp_input		
Description	Get LC<index> module<idx> temperature input		
Access	Read only		
Release version	V.7.0030.4000		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Get LC1 module1 temperature input: <code>cat \$bsp_path/thermal/lc1_module1_temp_input</code>		

LED Control

Common LED API reference: LED_Control_API.md — runtime led_<func> naming, sysfs tree, CPLD hardware ownership, worst-status aggregation, kernel patches, and examples. Runtime layout: examples/hw-management-led-sysfs.txt.

Get Fan Status LED

Node name	\$bsp_path/led/fan<index>_status_led		
Description	Get fan<index> status LED state		
Access	Read		
Release version	1.0		
Arguments	Name	Data type	Values
	fan	Integer	
Example	Get fan1 status LED: cat \$bsp_path/led/fan1_status_led		

Get Fan LED Capabilities

Node name	\$bsp_path/led/fan<index>_status_led_capability		
Description	Get fan<index> LED capabilities		
Access	Read		
Release version	1.0		
Arguments	Name	Data type	Values
	fan	Integer	
Example	Get fan1 LED capabilities: cat \$bsp_path/led/fan1_status_led_capability		

Set Fan LED Green/[Amber/Red]

Node name	\$bsp_path/led/fan<index>_status_led		
Description	Set fan<index> LED to green or amber/red		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	fan	Integer	
	color	String	green, amber, red
Example	Set fan1 LED to green: echo green > \$bsp_path/led/fan1_status_led		

Set Fan LED Green/[Amber/Red] Delay Off

Node name	\$bsp_path/led/fan<index>_status_led_delay_off		
Description	Set fan<index> LED delay off time		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	fan	Integer	
	delay	Integer	milliseconds
Example	Set fan1 LED delay off to 1000ms: echo 1000 > \$bsp_path/led/fan1_status_led_delay_off		

Set Fan LED Green/[Amber/Red] Delay On

Node name	\$bsp_path/led/fan<index>_status_led_delay_on		
Description	Set fan<index> LED delay on time		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	fan	Integer	
	delay	Integer	milliseconds
Example	Set fan1 LED delay on to 500ms: echo 500 > \$bsp_path/led/fan1_status_led_delay_on		

Get PSU LED

Node name	\$bsp_path/led/psu<index>_status_led		
Description	Get PSU<index> LED state		
Access	Read		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Get PSU1 LED: cat \$bsp_path/led/psu1_status_led		

Get PSU LED Capabilities

Node name	\$bsp_path/led/psu<index>_status_led_capability		
Description	Get PSU<index> LED capabilities		
Access	Read		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Get PSU1 LED capabilities: <code>cat \$bsp_path/led/psu1_status_led_capability</code>		

Set PSU LED Green/[Amber/Red]

Node name	\$bsp_path/led/psu<index>_status_led		
Description	Set PSU<index> LED to green or amber/red		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
	color	String	green, amber, red
Example	Set PSU1 LED to green: <code>echo green > \$bsp_path/led/psu1_status_led</code>		

Set PSU LED Green/[Amber/Red] Delay Off

Node name	\$bsp_path/led/psu<index>_status_led_delay_off		
Description	Set PSU<index> LED delay off time		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
	delay	Integer	milliseconds
Example	Set PSU1 LED delay off to 1000ms: echo 1000 > \$bsp_path/led/psu1_status_led_delay_off		

Set PSU LED Green/[Amber/Red] Delay On

Node name	\$bsp_path/led/psu<index>_status_led_delay_on		
Description	Set PSU<index> LED delay on time		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
	delay	Integer	milliseconds
Example	Set PSU1 LED delay on to 500ms: echo 500 > \$bsp_path/led/psu1_status_led_delay_on		

Get Status LED

Node name	\$bsp_path/led/status_led		
Description	Get system status LED state		
Access	Read		
Release version	1.0		
Arguments	Name	Data type	Values
	status	String	
Example	Get system status LED: cat \$bsp_path/led/status_led		

Get Status LED Capabilities

Node name	\$bsp_path/led/status_led_capability		
Description	Get system status LED capabilities		
Access	Read		
Release version	1.0		
Arguments	Name	Data type	Values
	capability	String	
Example	Get system status LED capabilities: cat \$bsp_path/led/status_led_capability		

Set Status Green/[Amber/Red]

Node name	\$bsp_path/led/status_led		
Description	Set system status LED to green or amber/red		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	color	String	green, amber, red
Example	Set system status LED to green: <code>echo green > \$bsp_path/led/status_led</code>		

Set Status LED Green/[Amber/Red] Delay Off

Node name	\$bsp_path/led/status_led_delay_off		
Description	Set system status LED delay off time		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	delay	Integer	milliseconds
Example	Set system status LED delay off to 1000ms: <code>echo 1000 > \$bsp_path/led/status_led_delay_off</code>		

Set Status LED Green/[Amber/Red] Delay On

Node name	\$bsp_path/led/status_led_delay_on		
Description	Set system status LED delay on time		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	delay	Integer	milliseconds
Example	Set system status LED delay on to 500ms: <code>echo 500 > \$bsp_path/led/status_led_delay_on</code>		

Get LED Control Owner

Stack	Host		
Node name	\$bsp_path/led/led_<name>_control		
Description	Control owner of LED <name> (status, fan, fan1, psu, uid, ...). Created on LED udev add next to led_<name>_capability and led_<name>_state. Value is resolved from \$bsp_path/config/led_control_type (§3.1.60). Missing map or unmatched name uses led_hw_sw. Value Meaning —— —— led_sw Software controlled led_hw Hardware controlled led_hw_sw Hardware and software controlled (default)		
Access	Read only		
Release version	3.2.7		
Arguments	Name	Data type	Values
	name	String	LED type from the udev LED class name
	control	String	led_sw, led_hw, led_hw_sw
Example	Get status LED control owner (default when no platform map): <code>cat \$bsp_path/led/led_status_control led_hw_sw</code>		

Power Control

Get Power Consumption

Node name	\$bsp_path/power/pwr_consum \$bsp_path/power/pwr_sys		
Description	Get system power consumption values - pwr_consum: Power consumption calculation based on PSU selection - pwr_sys: System power consumption		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu1	String	Get power consumption for PSU1
	psu2	String	Get power consumption for PSU2
Example	Get power consumption for PSU1: cat \$bsp_path/power/pwr_consum psu1		

Get PSU sensor Current + thresholds

Node name	<code>\$bsp_path/power/psu<index>_curr<idx>_input</code> <code>\$bsp_path/power/psu<index>_curr<idx>_crit</code> <code>\$bsp_path/power/psu<index>_curr<idx>_max</code>		
Description	Get PSU<index> current sensor readings and thresholds		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Get PSU1 current sensor: <code>cat \$bsp_path/power/psu1_curr1_input cat \$bsp_path/power/psu1_curr1_crit cat \$bsp_path/power/psu1_curr1_max</code>		

Get PSU sensor Voltage + thresholds

Node name	<code>\$bsp_path/power/psu<index>_in<idx>_input</code> <code>\$bsp_path/power/psu<index>_in<idx>_crit</code> <code>\$bsp_path/power/psu<index>_in<idx>_max</code>		
Description	Get PSU<index> voltage sensor readings and thresholds		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Get PSU1 voltage sensor: <code>cat \$bsp_path/power/psu1_in1_input cat \$bsp_path/power/psu1_in1_crit cat \$bsp_path/power/psu1_in1_max</code>		

Get PSU sensor Power + thresholds

Node name	\$bsp_path/power/psu<index>_power<idx>_input \$bsp_path/power/psu<index>_power<idx>_crit \$bsp_path/power/psu<index>_power<idx>_max		
Description	Get PSU<index> power sensor readings and thresholds		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Get PSU1 power sensor: cat \$bsp_path/power/psu1_power1_input cat \$bsp_path/power/psu1_power1_crit cat \$bsp_path/power/psu1_ power1_max		

Get PSU sensor capability

Node name	\$bsp_path/power/psu<index>_sensor_capability		
Description	Get PSU<index> sensor capabilities		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Get PSU1 sensor capabilities: cat \$bsp_path/power/psu1_sensor_ capability		

System / Power Control

Get ASIC Health

Node name	<code>\$bsp_path/system/asic_health</code> <code>\$bsp_path/system/asic<N>_health</code> (for multi-ASIC systems)		
Description	Get ASIC health status. For multi-ASIC systems (N61XX_LD family: N6100_LD), additional nodes exist for each ASIC. Note: SN6600_LD (SN66XX_LD) is a single-ASIC system; only <code>asic_health</code> applies (no <code>asic2_health..asic4_health</code> in that configuration). Note: N6100_LD has 4 ASICs with individual health monitoring: - <code>asic_health</code> (ASIC1), <code>asic2_health</code>, <code>asic3_health</code>, <code>asic4_health</code>		
Access	Read only		
Release version	1.0 (V.7.0060.1000 for multi-ASIC)		
Arguments	Name	Data type	Values
	health	Integer	0 (unhealthy), 1 (healthy)
Example	Get ASIC health: # Single-ASIC systems <code>cat \$bsp_path/system/asic_health</code> # Multi-ASIC systems (N6100_LD) <code>cat \$bsp_path/system/asic_health</code> # ASIC1 <code>cat \$bsp_path/system/asic2_health</code> # ASIC2 <code>cat \$bsp_path/system/asic3_health</code> # ASIC3 <code>cat \$bsp_path/system/asic4_health</code> # ASIC4		

MCU Reset Control

Node name	\$bsp_path/system/mcu<N>_reset		
Description	Control MCU (Microcontroller Unit) reset for systems with multiple MCUs. Used for managing embedded controllers. Note: This attribute is for systems with MCU reset support (N61XX_LD family: N6100_LD). N6100_LD has 2 MCUs (mcu1_reset, mcu2_reset).		
Access	Read/Write		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	N	Integer	1-2 (N6100_LD)
	reset	Integer	0 (normal), 1 (reset)
Example	Reset MCU1: echo 1 > \$bsp_path/system/mcu1_reset cat \$bsp_path/system/mcu1_reset cat \$bsp_path/system/mcu2_reset		

Get CPLD Major Version

Node name	\$bsp_path/system/cpld_major_version		
Description	Get CPLD major version		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	version	Integer	
Example	Get CPLD major version: cat \$bsp_path/system/cpld_major_version		

Get CPLD Part Number

Node name	\$bsp_path/system/cpld_part_number		
Description	Get CPLD part number		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	part_number	String	
Example	Get CPLD part number: cat \$bsp_path/system/cpld_part_number		

Get CPLD Minor Version

Node name	\$bsp_path/system/cpld_minor_version		
Description	Get CPLD minor version		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	version	Integer	
Example	Get CPLD minor version: cat \$bsp_path/system/cpld_minor_version		

Get CPLD Full Version

Node name	\$bsp_path/system/cpld_full_version		
Description	Get CPLD full version		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	version	String	
Example	Get CPLD full version: cat \$bsp_path/system/cpld_full_version		

Fan Direction

Node name	\$bsp_path/system/fan_direction		
Description	Get/set fan direction		
Access	Read/Write		
Release version	1.0		
Arguments	Name	Data type	Values
	direction	String	intake, exhaust
Example	Get fan direction: cat \$bsp_path/system/fan_direction		

Set JTAG Mode

Node name	\$bsp_path/system/jtag_mode		
Description	Set JTAG mode		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	mode	Integer	0/1
Example	Set JTAG mode: echo 1 > \$bsp_path/system/jtag_mode		

Set PSU On/Off

Node name	\$bsp_path/system/psu<index>_on_off		
Description	Set PSU<index> on/off		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
	state	Integer	0/1
Example	Set PSU1 on: echo 1 > \$bsp_path/system/psu1_on_off		

Set System Power Cycle

Node name	\$bsp_path/system/power_cycle		
Description	Set system power cycle		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	cycle	Integer	0/1
Example	Set system power cycle: echo 1 > \$bsp_path/system/power_cycle		

Set System Power Down

Node name	\$bsp_path/system/power_down		
Description	Set system power down		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	power	Integer	0/1
Example	Set system power down: echo 1 > \$bsp_path/system/power_down		

CPU Shut Down Request

Node name	\$bsp_path/system/cpu_shutdown_req		
Description	CPU initiated shutdown request signal. This attribute is used by the CPU to request a system shutdown sequence. On liquid-cooled platforms where hw_management_platform_config.py defines a monitor entry, hw-mgmt polls this node (not events/graceful_pwr_off) and invokes run_power_button_event when the value changes. Platforms with this entry in the current tree include HI162, HI166-HI170, HI176, HI177, HI180, HI185 (N6300_LD), and others listed in PLATFORM_CONFIG.		
Access	Read/Write		
Release version	1.0		
Arguments	Name	Data type	Values
	request	Integer	0/1
Example	Set CPU shutdown request: echo 1 > \$bsp_path/system/cpu_shutdown_req		

Graceful Power Off

Node name	\$bsp_path/system/graceful_power_off		
Description	Initiate graceful system power off sequence. This attribute triggers a controlled shutdown procedure that allows the system to properly close processes and save state before powering off.		
Access	Read/Write		
Release version	3.0		
Arguments	Name	Data type	Values
	poweroff	Integer	0/1
Example	Initiate graceful power off: echo 1 > \$bsp_path/system/graceful_power_off		
Note	See also section 3.5.9 for the related event graceful_pwr_off that monitors graceful shutdown status.		

CPU Power off Ready

Node name	\$bsp_path/system/cpu_power_off_ready		
Description	CPU signal indicating readiness for power off. This attribute is set by the CPU to signal that it has completed shutdown procedures and is ready for the system to power down.		
Access	Read/Write		
Release version	1.0		
Arguments	Name	Data type	Values
	ready	Integer	0/1
Example	Set CPU power off ready status: <code>echo 1 > \$bsp_path/system/cpu_power_off_ready</code>		

Shutdown Unlock

Node name	\$bsp_path/system/shutdown_unlock		
Description	Unlock mechanism for shutdown operations. This attribute must be set to enable shutdown and power cycle operations, providing a safety mechanism to prevent accidental system shutdowns.		
Access	Read/Write		
Release version	1.0		
Arguments	Name	Data type	Values
	unlock	Integer	0/1
Example	Unlock shutdown operations: <code>echo 1 > \$bsp_path/system/shutdown_unlock</code>		
Example	Lock shutdown operations: <code>echo 0 > \$bsp_path/system/shutdown_unlock</code>		

Boot Completed

Node name	\$bsp_path/system/boot_completed		
Description	System boot completion status. This attribute indicates that the system has completed its boot sequence and is fully operational.		
Access	Read/Write		
Release version	1.0		
Arguments	Name	Data type	Values
	status	Integer	0 (not completed), 1 (completed)
Example	Read boot completion status: <code>cat \$bsp_path/system/boot_completed</code>		
Example	Set boot completed flag: <code>echo 1 > \$bsp_path/system/boot_completed</code>		
Note	On AMD SNW CPU platforms, this may be implemented as a GPIO-based attribute.		

Set Line Card Power

Node name	\$bsp_path/system/lc<index>_power		
Description	Set line card<index> power		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	lc	Integer	
	power	Integer	0/1
Example	Set LC1 power: <code>echo 1 > \$bsp_path/system/lc1_power</code>		

Set Line Card Enable

Node name	\$bsp_path/system/lc<index>_enable		
Description	Set line card<index> enable		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	lc	Integer	
	enable	Integer	0/1
Example	Set LC1 enable: echo 1 > \$bsp_path/system/lc1_enable		

Read Line Card Active

Node name	\$bsp_path/system/lc<index>_active		
Description	Read line card<index> active status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Read LC1 active status: cat \$bsp_path/system/lc1_active		

Read Line Card Powered

Node name	\$bsp_path/system/lc<index>_powered		
Description	Read line card<index> powered status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Read LC1 powered status: cat \$bsp_path/system/lc1_powered		

Read Line Card Present

Node name	\$bsp_path/system/lc<index>_present		
Description	Read line card<index> present status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Read LC1 present status: cat \$bsp_path/system/lc1_present		

Read Line Card Ready

Node name	\$bsp_path/system/lc<index>_ready		
Description	Read line card<index> ready status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Read LC1 ready status: cat \$bsp_path/system/lc1_ready		

Read Line Card Synced

Node name	\$bsp_path/system/lc<index>_synced		
Description	Read line card<index> synced status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Read LC1 synced status: cat \$bsp_path/system/lc1_synced		

Read Line Card Verified

Node name	\$bsp_path/system/lc<index>_verified		
Description	Read line card<index> verified status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Read LC1 verified status: cat \$bsp_path/system/lc1_verified		

Read Line Card Reset Mask

Node name	\$bsp_path/system/lc<index>_reset_mask		
Description	Read line card<index> reset mask		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	lc	Integer	
Example	Read LC1 reset mask: cat \$bsp_path/system/lc1_reset_mask		

Set Line Card Shutdown

Node name	\$bsp_path/system/lc<index>_shutdown		
Description	Set line card<index> shutdown		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	lc	Integer	
	shutdown	Integer	0/1
Example	Set LC1 shutdown: echo 1 > \$bsp_path/system/lc1_shutdown		

Set VPD Write Protect

Node name	\$bsp_path/system/vpd_write_protect		
Description	Set VPD write protect		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	protect	Integer	0/1
Example	Set VPD write protect: echo 1 > \$bsp_path/system/vpd_write_protect		

Set ASIC Up during PCIe root complex reset

Node name	\$bsp_path/system/asic_up_during_pcie_reset		
Description	Set ASIC up during PCIe root complex reset		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	asic	Integer	0/1
Example	Set ASIC up during PCIe reset: echo 1 > \$bsp_path/system/asic_up_during_pcie_reset		

Get Voltreg Update status

Node name	\$bsp_path/system/voltreg_update_status		
Description	Get voltage regulator update status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	status	String	
Example	Get voltage regulator update status: cat \$bsp_path/system/voltreg_update_status		

Get Config1, Config2

Node name	\$bsp_path/system/config1 \$bsp_path/system/config2		
Description	Get system configuration 1 and 2		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	config	Integer	
Example	Get system configurations: cat \$bsp_path/system/config1 cat \$bsp_path/system/config2		

Get Ufm Version

Node name	\$bsp_path/system/ufm_version		
Description	Get UFM version		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	version	String	
Example	Get UFM version: cat \$bsp_path/system/ufm_version		

DPU Power Off

Node name	\$bsp_path/system/dpu<index>_power_off		
Description	Set DPU<index> power off		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	dpu	Integer	
	power	Integer	0/1
Example	Set DPU1 power off: echo 1 > \$bsp_path/system/dpu1_power_off		

DPU Force Power Off

Node name	\$bsp_path/system/dpu<index>_force_power_off		
Description	Force DPU<index> power off		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	dpu	Integer	
	force	Integer	0/1
Example	Force DPU1 power off: echo 1 > \$bsp_path/system/dpu1_force_power_off		

DPU Reset

Node name	\$bsp_path/system/dpu<index>_reset		
Description	Set DPU<index> reset		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	dpu	Integer	
	reset	Integer	0/1
Example	Set DPU1 reset: echo 1 > \$bsp_path/system/dpu1_reset		

DPU Reset Enable

Node name	\$bsp_path/system/dpu<index>_reset_enable		
Description	Enable DPU<index> reset		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	dpu	Integer	
	enable	Integer	0/1
Example	Enable DPU1 reset: echo 1 > \$bsp_path/system/dpu1_reset_enable		

Get Reset Cause

Node name	\$bsp_path/system/reset_<cause>																														
Description	Get last system reset cause. Reset causes vary between platform families. Each supported cause is exposed as a separate read-only attribute under \$bsp_path/system/. Reading \$bsp_path/system/reset_cause returns the primary cause string. For most causes only one attribute reads 1; the rest read 0.																														
Access	Read only																														
Release version	1.0																														
Arguments	Name	Data type	Values																												
	cause	String	Primary reset cause (via reset_cause)																												
	reset_<cause>	Integer	1 = this cause triggered the reset, 0 = not related																												
Example	Check if long button press caused reset: cat \$bsp_path/system/reset_long_pb																														
Example	Get reset cause string: cat \$bsp_path/system/reset_cause ##### N51XX_LD platform family Per CPLD design, N51XX_LD platforms do not support reset_ac_pwr_fail, reset_aux_pwr_or_ref, reset_from_asic, or reset_reload_bios. Those attributes must not be documented or tested as available on N51XX_LD systems. Applies to N5110_LD, N5112_LD, N5100_LD, N5101_LD, N5200_LD, N5201_LD, N5300_LD, N5400_LD, N5120_LD, N5121_LD, N5320_LD, N5500_LD (GB200/GB300), and N5240_LD (Kyber). config/reset_attr_num = 22. <table><tr><td> Reset cause attribute Description </td><td> </td><td> </td><td> reset_long_pb Reset button was pushed for more than 15 seconds </td></tr><tr><td> reset_short_pb Reset button was pushed for less than 15 seconds </td><td> </td><td> reset_aux_pwr_or_fu Reset due to CPLD power down or CPLD code refresh </td><td> </td></tr><tr><td> reset_swb_dc_dc_pwr_fail Switch board DC/DC power failure </td><td> </td><td> reset_sw_reset Power cycle command initiated by software </td><td> </td></tr><tr><td> reset_pwr_button_or_leak_con Reset triggered by power button or leak connector event </td><td> </td><td> reset_swb_wd Reset or power off initiated by switch-board watchdog </td><td> </td></tr><tr><td> reset_asic_thermal ASIC power drop due to failure or thermal shutdown </td><td> </td><td> reset_cpu_thermal CPU thermal shutdown </td><td> </td></tr><tr><td> reset_aux_pwr_or_reload Auxiliary power failure or CPLD field upgrade </td><td> </td><td> reset_comex_pwr_fail COMe module power failure </td><td> </td></tr><tr><td> reset_platform Reboot command initiated by software </td><td> </td><td> reset_soc Power off initiated by SOC (for example Linux poweroff) </td><td> </td></tr></table>			Reset cause attribute Description			reset_long_pb Reset button was pushed for more than 15 seconds	reset_short_pb Reset button was pushed for less than 15 seconds		reset_aux_pwr_or_fu Reset due to CPLD power down or CPLD code refresh		reset_swb_dc_dc_pwr_fail Switch board DC/DC power failure		reset_sw_reset Power cycle command initiated by software		reset_pwr_button_or_leak_con Reset triggered by power button or leak connector event		reset_swb_wd Reset or power off initiated by switch-board watchdog		reset_asic_thermal ASIC power drop due to failure or thermal shutdown		reset_cpu_thermal CPU thermal shutdown		reset_aux_pwr_or_reload Auxiliary power failure or CPLD field upgrade		reset_comex_pwr_fail COMe module power failure		reset_platform Reboot command initiated by software		reset_soc Power off initiated by SOC (for example Linux poweroff)	
Reset cause attribute Description			reset_long_pb Reset button was pushed for more than 15 seconds																												
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reset_platform Reboot command initiated by software		reset_soc Power off initiated by SOC (for example Linux poweroff)																													

Cartridge

Node name	\$bsp_path/system/cartridge<index>		
Description	Get cable cartridge status at the specified index. Cable cartridges are used in liquid-cooled multi-ASIC systems (N61XX_LD family: N6100_LD, N6300_LD) for connecting external cables. Note: This attribute is primarily for liquid-cooled systems with cable cartridges (N61XX_LD family: N6100_LD, N6300_LD). N6100_LD and N6300_LD (SKU HI185) have 4 cable cartridges.		
Access	Read only		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	index	Integer	1-4 (N6100_LD, N6300_LD)
	Status	Integer	0/1
Example	Get cartridge 1 status: cat \$bsp_path/system/cartridge1		

ASIC PG Failure

Node name	\$bsp_path/system/asic_pg_fail		
Description	Get ASIC power good failure status. Indicates power sequencing failure on the ASIC. Note: This attribute is for multi-ASIC systems (SN58XX_LD family, N61XX_LD family). N6100_LD has 4 ASICs with individual failure tracking. Exception: This API does not exist on SN6600_LD, SN6800_LD, and SN6810_LD systems.		
Access	Read only		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	Status	Integer	0/1
Example	Get ASIC power good failure status: <code>cat \$bsp_path/system/asic_pg_fail</code>		

Leakage Sensors

Leakage Sensor

Node name	\$bsp_path/system/leakage<index>		
Description	Get leakage sensor status at the specified index. Detects liquid leaks in liquid-cooled systems. Note: This attribute is for liquid-cooled systems (SN58XX_LD family, N61XX_LD family, SN66XX_LD family). N6100_LD has 2 leakage sensors, SN5810_LD has 2, SN5800_LD has 5. N6300_LD (SKU HI185): config/leakage_counter is 2 (n61xxld_specific() / HI185 branch in hw-management.sh). SN6600_LD: config/leakage_counter is 2; the validated sysfs tree may still list system/leakage1 through system/leakage5 when the hardware exposes those mlxreg-io channels (see the leakage_counter section note above).		
Access	Read only		
Release version	V.7.0040.3930		
Arguments	Name	Data type	Values
	index	Integer	1-X (system dependent)
	Status	Integer	0=no leak, 1=leak detected
Example	Get leakage sensor 1 status: cat \$bsp_path/system/leakage1		

Thermal

Reference — Host: usr/usr/bin/hw-management-thermal-events.sh; platform sensors in usr/etc/hw-management-sensors/.

Reference — BMC (HI189): bmc/usr/etc/HI189/hw-management-bmc-events.sh; bmc/examples/hw-management-bmc-thermal-sysfs.txt.

Ambient sensors

Stack	Host (hw-management-thermal-events.sh)		
Node name	\$bsp_path/thermal/<ambient_name> (platform-specific; examples include mng_amb, comex_amb, port_amb, fan_amb, swb_amb, cpu_amb)		
Description	Ambient temperature sensors are created by hw-management-thermal-events.sh when the corresponding hwmon device appears. The stable symlink name under \$bsp_path/thermal/ depends on the platform label and sensor type. Some platforms also expose management ambient via \$bsp_path/thermal/mng_amb (see MNG Temperature). Legacy documentation referenced \$bsp_path/system/amb_sens for mlxreg ambient enablement on certain systems.		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	millidegrees Celsius
Example	Read management ambient temperature (when present): cat \$bsp_path/thermal/mng_amb		

Read Switch ASIC Temperature

Node name	\$bsp_path/thermal/asic<index>_temp_input		
Description	Read switch ASIC<index> temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	asic	Integer	
Example	Read ASIC1 temperature: cat \$bsp_path/thermal/asic1_temp_input		

Read Switch ASIC Temperature Normal

Node name	\$bsp_path/thermal/asic<index>_temp_normal		
Description	Read switch ASIC<index> normal temperature threshold		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	asic	Integer	
Example	Read ASIC1 normal temperature: cat \$bsp_path/thermal/asic1_temp_normal		

Read Switch ASIC Temperature Critical

Node name	\$bsp_path/thermal/asic<index>_temp_crit		
Description	Read switch ASIC<index> critical temperature threshold		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	asic	Integer	
Example	Read ASIC1 critical temperature: cat \$bsp_path/thermal/asic1_temp_crit		

Read Switch ASIC Temperature Emergency

Node name	\$bsp_path/thermal/asic<index>_temp_emergency		
Description	Read switch ASIC<index> emergency temperature threshold		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	asic	Integer	
Example	Read ASIC1 emergency temperature: cat \$bsp_path/thermal/asic1_temp_emergency		

Read Switch ASIC Temperature Trip Critical

Node name	\$bsp_path/thermal/asic<index>_temp_trip_crit		
Description	Read switch ASIC<index> trip critical temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	asic	Integer	
Example	Read ASIC1 trip critical temperature: cat \$bsp_path/thermal/asic1_temp_trip_crit		

Read Switch Comex Temperature

Node name	\$bsp_path/thermal/comex_temp_input		
Description	Read switch Comex temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	
Example	Read Comex temperature: cat \$bsp_path/thermal/comex_temp_input		

MNG Temperature

Stack	Host (hw-management-thermal-events.sh)		
Node name	\$bsp_path/thermal/mng_amb		
Description	Read management (MNG) ambient temperature when the platform exposes this sensor through the hw-mgmt thermal events handler.		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	millidegrees Celsius
Example	Read MNG ambient temperature: cat \$bsp_path/thermal/mng_amb		

Read host CPU temperature (BMC stack)

Stack	BMC (hw-management-bmc-events.sh, event cpu_temp)
Node name	\$bsp_path/thermal/cpu_temp_input
Description	Instantaneous host CPU temperature as seen on the BMC image. The BMC early-I2C map binds bus 15 / 0x4c to the sbtsi driver (hw-management-bmc-early-i2c-devices.json). This is not created by the host thermal events script on the CPU image.
Access	Read only
Release version	1.0
Example	cat \$bsp_path/thermal/cpu_temp_input

Read host CPU temperature max / min (BMC stack)

Stack	BMC (hw-management-bmc-events.sh, event cpu_temp)
Node name	\$bsp_path/thermal/cpu_temp (maps to hwmon temp1_max); \$bsp_path/thermal/cpu_min (maps to hwmon temp1_min)
Description	High/low limit attributes from the same sbtsi sensor as cpu_temp_input. sbtsi exposes temp1_min; both symlinks are created on H1189.
Access	Read only
Release version	1.0
Example	cat \$bsp_path/thermal/cpu_temp cat \$bsp_path/thermal/cpu_min

Read BMC Temperature

Stack	BMC (hw-management-bmc-events.sh, event bmc_temp)		
Node name	\$bsp_path/thermal/bmc_temp_input		
Description	Read BMC ambient temperature (instantaneous reading from the BMC hwmon sensor). Symlink is created under /var/run/hw-management/thermal/ by hw-management-bmc-events.sh when the BMC temperature hwmon device appears (for example HI189 / SN6600: I2C 4-0048, 1m75 driver, temp1_input).		
Access	Read only		
Release version	1.0		
Note	Supported on systems with the hw-mgmt BMC thermal stack only. Values are typically millidegrees Celsius (Linux hwmon convention).		
Arguments	Name	Data type	Values
	temperature	Integer	millidegrees Celsius
Example	Read BMC ambient temperature: cat \$bsp_path/thermal/bmc_temp_input		

Read BMC Max Temperature

Stack	BMC (<code>hw-management-bmc-events.sh</code> , event <code>bmc_temp</code> — BMC board lm75 sensor, not host CPU)		
Node name	<code>\$bsp_path/thermal/bmc_temp</code>		
Description	Read BMC ambient high-temperature limit (maps to <code>hwmon temp1_max</code> on the BMC sensor). Created by the same BMC thermal udev handler as <code>bmc_temp_input</code> . On H1189 the BMC sensor uses the lm75 driver, which exposes <code>temp1_input</code> and <code>temp1_max</code> only (no <code>temp1_min</code>).		
Access	Read only		
Release version	1.0		
Note	Supported on systems with the hw-mgmt BMC thermal stack only. Do not expect <code>thermal/bmc_min</code> on lm75-backed BMC sensors; the driver does not register a minimum temperature attribute, so <code>hw-management-bmc-events.sh</code> does not create that symlink.		
Arguments	Name	Data type	Values
	temperature	Integer	driver-specific limit semantics
Example	Read BMC max temperature threshold: <code>cat \$bsp_path/thermal/bmc_temp</code>		

Read BMC Critical Temperature

Stack	BMC — not created on HI189 (lm75 has no temp1_crit)
Node name	<code>\$bsp_path/thermal/bmc_crit</code> (legacy); not created on HI189 BMC stack
Description	On systems with the hw-mgmt BMC thermal stack (HI189), the BMC ambient sensor uses the lm75 driver, which does not expose temp1_crit in sysfs. <code>hw-management-bmc-events.sh</code> does not create a <code>bmc_crit</code> symlink. Rev. 2.8 documented <code>\$bsp_path/thermal/bmc_crit</code> ; use <code>bmc_temp</code> (max/limit) and <code>bmc_temp_input</code> on current BMC platforms.
Access	Read only (when node exists on other platforms)
Release version	1.0
Note	Virtual test trees (for example HI193) may still ship fixture files named <code>bmc_crit</code> under <code>usr/etc/hw-management-virtual/</code> ; that naming does not apply to live HI189 runtime symlinks.
Example	Not applicable on HI189 — verify with <code>ls \$bsp_path/thermal/</code> on the device.

Read BMC Minimal Temperature

Stack	BMC — not created on HI189 (lm75 has no temp1_min; check_n_link skips)
Node name	<code>\$bsp_path/thermal/bmc_min</code> (legacy); not created on HI189 BMC stack
Description	The BMC events handler attempts <code>temp1_min</code> $\square$ <code>bmc_min</code> , but <code>check_n_link</code> silently skips missing sources. The HI189 BMC sensor (lm75 at I2C 4-0048) has no temp1_min attribute, so 'bmc_min' never appears on production SONiC images using this stack. See <code>bmc/examples/hw-management-bmc-thermal-sysfs.txt</code> .
Access	Read only (when node exists on other platforms)
Release version	1.0
Example	Not applicable on HI189: # Expected absent on lm75-backed BMC: <code>ls \$bsp_path/thermal/bmc_min</code>

Read Cooling State

Node name	\$bsp_path/thermal/cooling_state		
Description	Read system cooling state		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	state	String	
Example	Read cooling state: cat \$bsp_path/thermal/cooling_state		

Cooling Name

Node name	\$bsp_path/config/cooling_name		
Description	Thermal control cooling device name written by <code>hw-management-thermal-events.sh</code> during cooling device registration. Used by the thermal control daemon configuration.		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	name	String	platform-specific
Example	Read cooling name: cat \$bsp_path/config/cooling_name		

Read CPU Core Temperature

Node name	\$bsp_path/thermal/cpu_core<index>_temp_input		
Description	Read CPU core<index> temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	core	Integer	
Example	Read CPU core1 temperature: cat \$bsp_path/thermal/cpu_core1_temp_input		

CPU Core Critical Temperature

Node name	\$bsp_path/thermal/cpu_core<index>_temp_crit		
Description	Read CPU core<index> critical temperature threshold		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	core	Integer	
Example	Read CPU core1 critical temperature: cat \$bsp_path/thermal/cpu_core1_temp_crit		

CPU Core Critical Temperature Alarm

Node name	\$bsp_path/thermal/cpu_core<index>_temp_crit_alarm		
Description	Read CPU core<index> critical temperature alarm		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	core	Integer	
Example	Read CPU core1 critical temperature alarm: cat \$bsp_path/thermal/cpu_core1_temp_crit_alarm		

CPU Core Temperature Max

Node name	\$bsp_path/thermal/cpu_core<index>_temp_max		
Description	Read CPU core<index> maximum temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	core	Integer	
Example	Read CPU core1 maximum temperature: cat \$bsp_path/thermal/cpu_core1_temp_max		

Read CPU Pack Temperature

Node name	\$bsp_path/thermal/cpu_pack_temp_input		
Description	Read CPU pack temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	
Example	Read CPU pack temperature: cat \$bsp_path/thermal/cpu_pack_temp_input		

CPU Pack Critical Temperature

Node name	\$bsp_path/thermal/cpu_pack_temp_crit		
Description	Read CPU pack critical temperature threshold		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	
Example	Read CPU pack critical temperature: cat \$bsp_path/thermal/cpu_pack_temp_crit		

CPU Pack Critical Temperature Alarm

Node name	\$bsp_path/thermal/cpu_pack_temp_crit_alarm		
Description	Read CPU pack critical temperature alarm		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	alarm	Integer	
Example	Read CPU pack critical temperature alarm: cat \$bsp_path/thermal/cpu_pack_temp_crit_alarm		

CPU Pack Temperature Max

Node name	\$bsp_path/thermal/cpu_pack_temp_max		
Description	Read CPU pack maximum temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	
Example	Read CPU pack maximum temperature: cat \$bsp_path/thermal/cpu_pack_temp_max		

Read Fan Max Speed

Node name	\$bsp_path/thermal/fan<index>_max_speed		
Description	Read fan<index> maximum speed		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	fan	Integer	
Example	Read fan1 maximum speed: cat \$bsp_path/thermal/fan1_max_speed		

Read Fan Min Speed

Node name	\$bsp_path/thermal/fan<index>_min_speed		
Description	Read fan<index> minimum speed		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	fan	Integer	
Example	Read fan1 minimum speed: cat \$bsp_path/thermal/fan1_min_speed		

Set Fan Speed

Node name	\$bsp_path/thermal/fan<index>_speed_set		
Description	Set fan PWM/speed for fan <index>. Symlink is created by <code>hw-management-thermal-events.sh</code> from the cooling device <code>pwm1</code> attribute when supported.		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	speed	Integer	platform-specific PWM scale
Example	Set fan1 speed: <code>echo 50 > \$bsp_path/thermal/fan1_speed_set</code>		

Fan Speed Tolerance

Node name	\$bsp_path/thermal/fan<index>_speed_tolerance		
Description	Fan speed tolerance for fan <index> when exposed under the thermal hierarchy. For system-wide tolerance configuration see also <code>\$bsp_path/config/fan_speed_tolerance</code> (§3.1.24).		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	tolerance	Integer	
Example	Read fan1 speed tolerance (when present): <code>cat \$bsp_path/thermal/fan1_speed_tolerance</code>		

Read Fan Direction

Node name	\$bsp_path/thermal/fan<index>_dir
Description	Read fan drawer <index> airflow direction.
Access	Read only
Release version	1.0
	Values: Value Meaning —— ——— 0 Reverse (C2P) 1 Forward (P2C) 2 Direction cannot be determined (fan missing, or fan present but direction debounce failed)
	Notes: - If the fan is not present (thermal/fan<index>_status is 0), fan<index>_dir is irrelevant and must be ignored by consumers. - On systems that report direction via CPLD (system/fan_dir), a remove event keeps fan<index>_dir present and sets it to 2. - On systems with fan-module EEPROM direction detection (config/fan_dir_eeprom), direction is taken from EEPROM VPD data. When the fan (and its EEPROM) is removed, fan<index>_dir may also be removed.
Example	Read fan1 direction: cat \$bsp_path/thermal/fan1_dir

Read Fan Status

Node name	\$bsp_path/thermal/fan<index>_status		
Description	Read fan<index> status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	fan	Integer	
Example	Read fan1 status: cat \$bsp_path/thermal/fan1_status		

Read Fan Fault

Node name	\$bsp_path/thermal/fan<index>_fault		
Description	Read fan<index> fault status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	fan	Integer	
Example	Read fan1 fault status: cat \$bsp_path/thermal/fan1_fault		

Comex Voltmon Temperature

Node name	\$bsp_path/thermal/comex_voltmon<index>_temp<index>_input		
Description	Read Comex PMBus voltmon temperature input.		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	millidegrees Celsius
Example	Read comex voltmon1 temperature: cat \$bsp_path/thermal/comex_voltmon1_temp1_input		

Comex Voltmon Critical Temperature

Node name	\$bsp_path/thermal/comex_voltmon<index>_temp<index>_crit		
Description	Read Comex voltmon critical temperature threshold.		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	millidegrees Celsius
Example	cat \$bsp_path/thermal/comex_voltmon1_temp1_crit		

Comex Voltmon Max Temperature

Node name	\$bsp_path/thermal/comex_voltmon<index>_temp<index>_max		
Description	Read Comex voltmon maximum temperature threshold.		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	millidegrees Celsius
Example	cat \$bsp_path/thermal/comex_voltmon1_temp1_max		

Read Port Ambient

Node name	\$bsp_path/thermal/port<index>_ambient_temp		
Description	Read port<index> ambient temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	port	Integer	
Example	Read port1 ambient temperature: cat \$bsp_path/thermal/port1_ambient_temp		

Read PSU Temperature

Node name	\$bsp_path/thermal/psu<index>_temp_input		
Description	Read PSU<index> temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Read PSU1 temperature: cat \$bsp_path/thermal/psu1_temp_input		

Read PSU Alarm

Node name	\$bsp_path/thermal/psu<index>_alarm		
Description	Read PSU<index> alarm status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Read PSU1 alarm: cat \$bsp_path/thermal/psu1_alarm		

Read PSU Max

Node name	\$bsp_path/thermal/psu<index>_temp_max		
Description	Read PSU<index> maximum temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Read PSU1 maximum temperature: cat \$bsp_path/thermal/psu1_temp_max		

Read PSU Fan Speed

Node name	\$bsp_path/thermal/psu<index>_fan_speed		
Description	Read PSU<index> fan speed		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Read PSU1 fan speed: cat \$bsp_path/thermal/psu1_fan_speed		

Read PSU min/max Fan Speed

Node name	\$bsp_path/thermal/psu<index>_fan_min_speed \$bsp_path/thermal/psu<index>_fan_max_speed		
Description	Read PSU<index> minimum and maximum fan speed		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Read PSU1 fan speeds: cat \$bsp_path/thermal/psu1_fan_min_speed cat \$bsp_path/thermal/psu1_fan_max_speed		

Read PSU Power Status

Node name	\$bsp_path/thermal/psu<index>_power_status		
Description	Read PSU<index> power status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Read PSU1 power status: cat \$bsp_path/thermal/psu1_power_status		

Read PSU Status

Node name	\$bsp_path/thermal/psu<index>_status		
Description	Read PSU<index> status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Read PSU1 status: cat \$bsp_path/thermal/psu1_status		

Read PDB Hotswap Controller Temperature

Node name	\$bsp_path/thermal/pdb_hotswap<index>_temp<index>_input		
Description	Read PDB (Power Distribution Board) hot-swap controller temperature Note: This attribute is for liquid-cooled systems only (SN58XX_LD family: SN5810_LD, SN5800_LD; N61XX_LD family: N6100_LD; SN66XX_LD family: SN6600_LD)		
Access	Read only		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Temperature	Integer	Value in millidegrees Celsius (m°C)
Example	Read PDB hotswap 1 temperature: cat \$bsp_path/thermal/pdb_hotswap1_temp1_input		

Read PDB Hotswap Controller Temperature Thresholds

Node name	\$bsp_path/thermal/pdb_hotswap<index>_temp<index>_crit \$bsp_path/thermal/pdb_hotswap<index>_temp<index>_max		
Description	Read PDB hot-swap controller temperature critical and maximum thresholds Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)		
Access	Read only		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Temperature	Integer	Value in millidegrees Celsius (m°C)
Example	Read PDB hotswap 1 temperature thresholds: cat \$bsp_path/thermal/pdb_hotswap1_temp1_crit cat \$bsp_path/thermal/pdb_hotswap1_temp1_max		

Read PDB Power Converter Temperature

Node name	\$bsp_path/thermal/pdb_pwr_conv<index>_temp<index>_input		
Description	Read PDB power converter temperature Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)		
Access	Read only		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Temperature	Integer	Value in millidegrees Celsius (m°C)
Example	Read PDB power converter 1 temperature: cat \$bsp_path/thermal/pdb_pwr_conv1_temp1_input		

Read PDB Power Converter Temperature Thresholds

Node name	\$bsp_path/thermal/pdb_pwr_conv<index>_temp<index>_crit \$bsp_path/thermal/pdb_pwr_conv<index>_temp<index>_lcrit \$bsp_path/thermal/pdb_pwr_conv<index>_temp<index>_max		
Description	Read PDB power converter temperature thresholds (critical, lower critical, maximum) Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)		
Access	Read only		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Temperature	Integer	Value in millidegrees Celsius (m°C)
Example	Read PDB power converter 1 temperature thresholds: cat \$bsp_path/thermal/pdb_pwr_conv1_temp1_crit cat \$bsp_path/thermal/pdb_pwr_conv1_temp1_lcrit cat \$bsp_path/thermal/pdb_pwr_conv1_temp1_max		

Read PDB MOSFET Ambient Temperature

Node name	\$bsp_path/thermal/pdb_mosfet_amb<index>		
Description	Read PDB MOSFET ambient temperature sensor Note: This attribute is for liquid-cooled systems only (SN58XX_LD family, N61XX_LD family, SN66XX_LD family)		
Access	Read only		
Release version	V.7.0050.3000		
Arguments	Name	Data type	Values
	Temperature	Integer	Value in millidegrees Celsius (m°C)
Example	Read PDB MOSFET ambient temperature: cat \$bsp_path/thermal/pdb_mosfet_amb1		

Read System PWM1

Node name	\$bsp_path/thermal/system_pwm1		
Description	Read system PWM1 value		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	pwm	Integer	
Example	Read system PWM1: cat \$bsp_path/thermal/system_pwm1		

Read Temperature Critical Module

Node name	\$bsp_path/thermal/module<index>_temp_crit		
Description	Read module<index> critical temperature threshold		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	module	Integer	
Example	Read module1 critical temperature: cat \$bsp_path/thermal/module1_temp_crit		

Read Temperature Emergency Module

Node name	\$bsp_path/thermal/module<index>_temp_emergency		
Description	Read module<index> emergency temperature threshold		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	module	Integer	
Example	Read module1 emergency temperature: cat \$bsp_path/thermal/module1_temp_emergency		

Read Temperature Trip Critical Module

Node name	\$bsp_path/thermal/module<index>_temp_trip_crit		
Description	Read module<index> trip critical temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	module	Integer	
Example	Read module1 trip critical temperature: cat \$bsp_path/thermal/module1_temp_trip_crit		

Read Temperature Fault Module

Node name	\$bsp_path/thermal/module<index>_temp_fault		
Description	Read module<index> temperature fault status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	module	Integer	
Example	Read module1 temperature fault: <code>cat \$bsp_path/thermal/module1_temp_fault</code>		

Read Temperature Input Module

Node name	\$bsp_path/thermal/module<index>_temp_input		
Description	Read module<index> temperature input		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	module	Integer	
Example	Read module1 temperature input: <code>cat \$bsp_path/thermal/module1_temp_input</code>		

Read Temperature Critical Gearbox

Node name	\$bsp_path/thermal/gearbox<index>_temp_crit		
Description	Read gearbox<index> critical temperature threshold		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	gearbox	Integer	
Example	Read gearbox1 critical temperature: cat \$bsp_path/thermal/gearbox1_temp_crit		

Read Temperature Emergency Gearbox

Node name	\$bsp_path/thermal/gearbox<index>_temp_emergency		
Description	Read gearbox<index> emergency temperature threshold		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	gearbox	Integer	
Example	Read gearbox1 emergency temperature: cat \$bsp_path/thermal/gearbox1_temp_emergency		

Read Temperature Trip Critical Gearbox

Node name	\$bsp_path/thermal/gearbox<index>_temp_trip_crit		
Description	Read gearbox<index> trip critical temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	gearbox	Integer	
Example	Read gearbox1 trip critical temperature: <code>cat \$bsp_path/thermal/gearbox1_temp_trip_crit</code>		

Read Temperature Input Gearbox

Node name	\$bsp_path/thermal/gearbox<index>_temp_input		
Description	Read gearbox<index> temperature input		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	gearbox	Integer	
Example	Read gearbox1 temperature input: <code>cat \$bsp_path/thermal/gearbox1_temp_input</code>		

Read Switch CPU Temperature

Node name	\$bsp_path/thermal/switch_cpu_temp		
Description	Read switch CPU temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	
Example	Read switch CPU temperature: cat \$bsp_path/thermal/switch_cpu_temp		

Read Switch Fan Temperature

Node name	\$bsp_path/thermal/switch_fan_temp		
Description	Read switch fan temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	
Example	Read switch fan temperature: cat \$bsp_path/thermal/switch_fan_temp		

Read Switch Port Temperature

Node name	\$bsp_path/thermal/switch_port<index>_temp		
Description	Read switch port<index> temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	port	Integer	
Example	Read switch port1 temperature: cat \$bsp_path/thermal/switch_port1_temp		

Read Switch Power Supply Temperature

Node name	\$bsp_path/thermal/switch_psu<index>_temp		
Description	Read switch power supply<index> temperature		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	psu	Integer	
Example	Read switch PSU1 temperature: cat \$bsp_path/thermal/switch_psu1_temp		

SODIMM Temperature Input

Node name	\$bsp_path/thermal/sodimm<index>_temp_input		
Description	Read SODIMM (Small Outline Dual In-line Memory Module) temperature. SODIMMs are DDR memory modules with integrated temperature sensors. Note: This attribute is for systems with SODIMM temperature sensors (SN58XX_LD family, N61XX_LD family, SN66XX_LD family). SN5810_LD and N6100_LD use 2 SODIMM sensors on I2C bus 2 (addresses 0x1a, 0x1b). SN6600_LD uses 2 JC42 sensors on I2C bus 10 (addresses 0x52, 0x53) per validated tree.		
Access	Read only		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	index	Integer	1-2 (N6100_LD, SN6600_LD)
	Temperature	Integer	Value in millidegrees Celsius (m°C)
Example	Read SODIMM temperatures: cat \$bsp_path/thermal/sodimm1_temp_input # SODIMM 1 cat \$bsp_path/thermal/sodimm2_temp_input # SODIMM 2		

SODIMM Critical Temperature

Node name	\$bsp_path/thermal/sodimm<index>_temp_crit		
Description	Read SODIMM critical temperature threshold. Note: This attribute is for systems with SODIMM temperature sensors (SN58XX_LD family, N61XX_LD family, SN66XX_LD family).		
Access	Read only		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	index	Integer	1-2 (N6100_LD, SN6600_LD)
	Temperature	Integer	Value in millidegrees Celsius (m°C)
Example	Read SODIMM critical temperatures: cat \$bsp_path/thermal/sodimm1_temp_crit cat \$bsp_path/thermal/sodimm2_temp_crit		

SODIMM Max Temperature

Node name	\$bsp_path/thermal/sodimm<index>_temp_max		
Description	Read SODIMM maximum temperature threshold. Note: This attribute is for systems with SODIMM temperature sensors (SN58XX_LD family, N61XX_LD family, SN66XX_LD family).		
Access	Read only		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	index	Integer	1-2 (N6100_LD, SN6600_LD)
	Temperature	Integer	Value in millidegrees Celsius (m°C)
Example	Read SODIMM max temperatures: cat \$bsp_path/thermal/sodimm1_temp_max cat \$bsp_path/thermal/sodimm2_temp_max		

SODIMM Min Temperature

Node name	\$bsp_path/thermal/sodimm<index>_temp_min		
Description	Read SODIMM minimum temperature threshold. Note: This attribute is for systems with SODIMM temperature sensors (SN58XX_LD family, N61XX_LD family, SN66XX_LD family).		
Access	Read only		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	index	Integer	1-2 (N6100_LD, SN6600_LD)
	Temperature	Integer	Value in millidegrees Celsius (m°C)
Example	Read SODIMM min temperatures: <code>cat \$bsp_path/thermal/sodimm1_temp_min cat \$bsp_path/thermal/sodimm2_temp_min</code>		

SODIMM Temperature Alarms

Node name	\$bsp_path/thermal/sodimm<index>_temp_crit_alarm \$bsp_path/thermal/sodimm<index>_temp_max_alarm \$bsp_path/thermal/sodimm<index>_temp_min_alarm		
Description	Read SODIMM temperature alarm status (critical, maximum, minimum). Note: This attribute is for systems with SODIMM temperature sensors (SN58XX_LD family, N61XX_LD family, SN66XX_LD family).		
Access	Read only		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	index	Integer	1-2 (N6100_LD, SN6600_LD)
	Alarm	Integer	0 (clear), 1 (alarm)
Example	Read SODIMM temperature alarms: cat \$bsp_path/thermal/sodimm1_temp_crit_alarm cat \$bsp_path/thermal/sodimm1_temp_max_alarm cat \$bsp_path/thermal/sodimm1_temp_min_alarm cat \$bsp_path/thermal/sodimm2_temp_crit_alarm cat \$bsp_path/thermal/sodimm2_temp_max_alarm cat \$bsp_path/thermal/sodimm2_temp_min_alarm		

SODIMM Temperature Hysteresis

Node name	<code>\$bsp_path/thermal/sodimm<index>_temp_crit_hyst</code> <code>\$bsp_path/thermal/sodimm<index>_temp_max_hyst</code>		
Description	Read SODIMM temperature hysteresis values for critical and max thresholds. Note: This attribute is for systems with SODIMM temperature sensors (SN58XX_LD family, N61XX_LD family, SN66XX_LD family).		
Access	Read only		
Release version	V.7.0060.1000		
Arguments	Name	Data type	Values
	index	Integer	1-2 (N6100_LD, SN6600_LD)
	Hysteresis	Integer	Value in millidegrees Celsius (m°C)
Example	Read SODIMM temperature hysteresis: <code>cat \$bsp_path/thermal/sodimm1_temp_crit_hyst</code> <code>cat \$bsp_path/thermal/sodimm1_temp_max_hyst</code> <code>cat \$bsp_path/thermal/sodimm2_temp_crit_hyst</code> <code>cat \$bsp_path/thermal/sodimm2_temp_max_hyst</code>		

SWB ASIC Temperature

Node name	<code>\$bsp_path/thermal/swb_asic<index></code>		
Description	Read switch-board (SWB) ASIC temperature when present on multi-board systems.		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	millidegrees Celsius
Example	<code>cat \$bsp_path/thermal/swb_asic1</code>		

Drive Temperature

Node name	\$bsp_path/thermal/drivetemp		
Description	Read NVMe/drive temperature from the drivetemp hwmon driver when linked by hw-management-thermal-events.sh.		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	temperature	Integer	millidegrees Celsius
Example	cat \$bsp_path/thermal/drivetemp		

Drive Critical Temperature

Node name	\$bsp_path/thermal/drivetemp_crit		
Description	Read drive critical temperature threshold.		
Access	Read only		
Release version	1.0		
Example	cat \$bsp_path/thermal/drivetemp_crit		

Drive Maximum Temperature Threshold

Node name	\$bsp_path/thermal/drivetemp_max		
Description	Read drive maximum temperature threshold.		
Access	Read only		
Release version	1.0		
Example	cat \$bsp_path/thermal/drivetemp_max		

Drive Minimum Temperature Threshold

Node name	<code>\$bsp_path/thermal/drivetemp_min</code>
Description	Read drive minimum temperature threshold.
Access	Read only
Release version	1.0
Example	<code>cat \$bsp_path/thermal/drivetemp_min</code>

Watchdog

Note: On SN6600_LD the validated hierarchy places attributes under `$bsp_path/watchdog/main/` and `$bsp_path/watchdog/aux/` instead of directly under `$bsp_path/watchdog/`.

Read Boot Status

Node name	<code>\$bsp_path/watchdog/bootstatus</code>		
Description	Read watchdog boot status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	status	Integer	
Example	Read watchdog boot status: <code>cat \$bsp_path/watchdog/bootstatus</code>		

Read Identity

Node name	\$bsp_path/watchdog/identity		
Description	Read watchdog identity		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	identity	String	
Example	Read watchdog identity: cat \$bsp_path/watchdog/identity		

Read No Way Out

Node name	\$bsp_path/watchdog/nowayout		
Description	Read watchdog no way out setting		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	nowayout	Integer	0/1
Example	Read watchdog no way out: cat \$bsp_path/watchdog/nowayout		

Read State

Node name	\$bsp_path/watchdog/state		
Description	Read watchdog state		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	state	String	
Example	Read watchdog state: cat \$bsp_path/watchdog/state		

Read Status

Node name	\$bsp_path/watchdog/status		
Description	Read watchdog status		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	status	String	
Example	Read watchdog status: cat \$bsp_path/watchdog/status		

Read Timeout

Node name	\$bsp_path/watchdog/timeout		
Description	Read watchdog timeout value		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	timeout	Integer	seconds
Example	Read watchdog timeout: cat \$bsp_path/watchdog/timeout		

Read Timeleft

Node name	\$bsp_path/watchdog/timeleft		
Description	Read watchdog time left		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	timeleft	Integer	seconds
Example	Read watchdog time left: cat \$bsp_path/watchdog/timeleft		

BMC status

Sections §3.23.1–§3.23.2 describe **host-CPU** views of BMC presence and I2C ownership (`hw-management.sh` / platform GPIO and `mlxreg-io`). §3.23.3–§3.23.5 are **host** config/status for MCTP setup where the platform defines it (`hw-management.sh`).

On the **BMC** image, `bmc_to_cpu_ctrl` also appears under `$bsp_path/system/` via the **mlxreg-io** regio handler (see `bmc/examples/hw-management-bmc-system-sysfs.txt`).

BMC present

Stack	Host
Node name	<code>\$bsp_path/system/bmc_present</code>
Description	GPIO indicating whether a BMC is present. Created by host <code>hw-management.sh</code> (platform-specific GPIO mapping).
Access	Read only
Release version	1.0
Example	<code>cat \$bsp_path/system/bmc_present</code>

BMC to CPU control

Stack	Host (mlxreg-io on CPU); also BMC (system/ symlink from regio on BMC image)
Node name	<code>\$bsp_path/system/bmc_to_cpu_ctrl</code>
Description	I2C bus ownership between BMC and CPU (0 = CPU, 1 = BMC on supported platforms). On the host image the symlink targets mlxplat <code>mlxreg-io</code> ; thermal control may read this node (<code>hw_management_thermal_control.py: system/bmc_to_cpu_ctrl</code>).
Access	Read/Write (platform-dependent)
Release version	1.0
Example	<code>cat \$bsp_path/system/bmc_to_cpu_ctrl</code>

MCTP address

Stack	Host (<code>hw-management.sh</code> writes when platform defines MCTP, e.g. N5110)
Node name	<code>\$bsp_path/config/mctp_addr</code>

Description	MCTP I2C client address used during host init to instantiate the MCTP device on <code>mctp_bus</code> .
Access	Read only (after init)
Release version	1.0
Example	<code>cat \$bsp_path/config/mctp_addr</code>

MCTP bus

Stack	Host
Node name	<code>\$bsp_path/config/mctp_bus</code>

Description	I2C bus number for MCTP client binding during host init.
Access	Read only (after init)
Release version	1.0
Example	<code>cat \$bsp_path/config/mctp_bus</code>

MCTP ready

Stack	Host (mlxreg-io); BMC register <code>cpu_mctp_ready</code> under <code>\$bsp_path/system/</code> on BMC image
Node name	<code>\$bsp_path/system/cpu_mctp_ready</code>
Description	Indicates CPU MCTP readiness from platform mlxreg-io. Present in validated host trees (SN6600_LD, SN5810_LD, N6100_LD). On the BMC image the same logical register is listed in <code>nvsb_bmc_hid189_regio_data / hw-management-bmc-system-sysfs.txt</code> .
Access	Read only
Release version	1.0
Example	<code>cat \$bsp_path/system/cpu_mctp_ready</code>

BMC reset cause

Stack: BMC (`hw-management-bmc` package on AST2700 SONiC BMC OS)

The BMC reset-cause exporter (`hw-management-bmc-get-reset-cause.sh`) writes a separate subtree under `$bsp_path/bmc/`. This is distinct from host CPLD reset causes documented in §3.18.39 (`$bsp_path/system/reset_*` on the switch CPU).

Source priority for SCU register words: U-Boot environment (`fw_printenv`), then `/proc/cmdline`, then `devmem`. Operator summary: `hw-management-bmc-show-reset-cause.sh` (see `bmc/README.md`).

Validation source: `bmc/usr/bin/hw-management-bmc-get-reset-cause.sh`, `bmc/examples/hw-management-bmc-system-sysfs.txt` (host-side reset attrs only).

Primary BMC reset cause

Stack: BMC

Exactly **one** of the following files reads **1**; the others read **0** (v2 primary heuristic for H1189 / AST2700).

Node name	Description
<code>\$bsp_path/bmc/reset_pwr_cycle</code>	Power-cycle-like BMC boot (no WDT log; SCU0 EXTRST# set)
<code>\$bsp_path/bmc/reset_soft_reboot</code>	Warm reboot (WDT logged, or no WDT and EXTRST# clear)
<code>\$bsp_path/bmc/reset_unknown</code>	WDT log and EXTRST# both set (conflicting / sticky path)

Access: Read only

Release version: V.7.0070.1000

Example:

```
cat $bsp_path/bmc/reset_pwr_cycle
cat $bsp_path/bmc/reset_soft_reboot
cat $bsp_path/bmc/reset_unknown
```

BMC reset domain detail

Stack: BMC

Hardware domain flags decoded from AST2700 SCU reset-event logs. Each file reads **0** or **1**.

Node name	Description
<code>\$bsp_path/bmc/domains/reset_power_on</code>	PWRST indication (SCU0/SCU1)
<code>\$bsp_path/bmc/domains/reset_external</code>	EXTRST# / SRST evidence
<code>\$bsp_path/bmc/domains/reset_watchdog</code>	Non-software WDT bits or SCU0 watchdog evidence
<code>\$bsp_path/bmc/domains/reset_software</code>	Software WDT bits in SCU1 0x080

Node name	Description
<code>\$bsp_path/bmc/domains/reset_cpu</code>	CPU domain reset
<code>\$bsp_path/bmc/domains/reset_soc</code>	SoC domain reset
<code>\$bsp_path/bmc/domains/reset_ahb</code>	AHB domain reset
<code>\$bsp_path/bmc/domains/reset_caliptra</code>	Caliptra domain reset
<code>\$bsp_path/bmc/domains/reset_usb</code>	USB subsystem reset
<code>\$bsp_path/bmc/domains/reset_spi</code>	SPI domain reset
<code>\$bsp_path/bmc/domains/reset_espi</code>	eSPI domain reset
<code>\$bsp_path/bmc/domains/reset_emmc</code>	eMMC reset (SCU0)
<code>\$bsp_path/bmc/domains/reset_msi</code>	MSI reset (SCU0)
<code>\$bsp_path/bmc/domains/reset_security_watchdog2</code>	WDT2 nibble in SCU1 0x080
<code>\$bsp_path/bmc/domains/reset_others</code>	No PWRST/WDT/CPU/WDT2 bits set

Access: Read only

Release version: V.7.0070.1000

Example:

```
cat $bsp_path/bmc/domains/reset_watchdog
```

Raw SCU reset event logs

Stack: BMC

Raw 32-bit SCU reset-event register values (hex text), written at export time.

Node name	SCU source (AST2700)
<code>\$bsp_path/bmc/raw_scu0_reset_event_log0</code>	SCU0 0x050
<code>\$bsp_path/bmc/raw_scu0_reset_event_log2</code>	SCU0 0x070

Node name	SCU source (AST2700)
<code>\$bsp_path/bmc/raw_scu1_reset_event_log0</code>	SCU1 0x050
<code>\$bsp_path/bmc/raw_scu1_reset_event_log3</code>	SCU1 0x080

Access: Read only

Release version: V.7.0070.1000

Example:

```
cat $bsp_path/bmc/raw_scu0_reset_event_log0
```

BMC leakage A2D tree

Stack: BMC

Runtime leak-detector hierarchy under `$bsp_path/leakage/`, populated by `hw-management-bmc-a2d-leakage-config.sh` from `/etc/hw-management-bmc-a2d-leakage-config.json` (see `bmc/examples/hw-management-bmc-a2d-leakage-config-example.json`). Supported device types in the example config include **MAX1363**, **ADS1015**, and **ADS7924**. Udev events are handled by `hw-management-bmc-leakage-handler.sh`.

Layout reference: `bmc/examples/hw-management-bmc-leakage-sysfs.txt`.

N = leak-detector index (1-based order in the JSON array). **j** = channel index; channel directory names follow hardware `Channels[]`. `Id` and may be non-contiguous (for example `.../1/` and `.../4/` only).

This tree is separate from host `$bsp_path/system/leakage<N> mlxreg-io` status (§3.19.1) and from `$bsp_path/events/leakage<N>` on the CPU image.

Leak detector attributes

Stack: BMC

Node name	Description
<code>\$bsp_path/leakage/<N>/device_type</code>	Device type from JSON (for example ADS7924)
<code>\$bsp_path/leakage/<N>/device_name</code>	Detector name from JSON Name field

Access: Read only

Release version: V.7.0070.1000

Example:

```
cat $bsp_path/leakage/1/device_type
cat $bsp_path/leakage/1/device_name
```

Leak channel attributes

Stack: BMC

Per-channel directory: `$bsp_path/leakage/<N>/<j>/`

Node name	Description
<code>input</code>	Raw IIO sample (symlink or file when Probe is true in JSON)
<code>min</code>	Low threshold (register-derived × scale, config units)
<code>max</code>	High threshold (register-derived × scale)
<code>warn</code>	WarningMax for this channel's Type
<code>crit</code>	CriticalMax for this channel's Type
<code>lwarn</code>	Optional WarningMin for this channel's Type
<code>lcrit</code>	Optional CriticalMin for this channel's Type
<code>type</code>	Channel type from JSON: rop, flex, or embedded
<code>scale</code>	Scale factor from JSON

Node name	Description
<code>channel_name</code>	Optional human-readable name from JSON ChnlNames
<code>last_sample</code>	Optional 12-bit-aligned code (written on OOB event by leakage handler)
<code>last_event</code>	Optional event timestamp in milliseconds (written by leakage handler)

Access: Read only (handler may write `last_sample` / `last_event` on out-of-band events)

Release version: V.7.0070.1000

Example:

```
cat $bsp_path/leakage/1/1/input
cat $bsp_path/leakage/1/1/warn
cat $bsp_path/leakage/1/1/type
```

Note: On BMC images for H1189, `mlxreg-hotplug` also exposes aggregate status registers such as `$bsp_path/system/leakage1`, `$bsp_path/system/leakage2`, and `$bsp_path/system/leakage_aggr` (See `bmc/examples/hw-management-bmc-system-sysfs.txt`).

JTAG interface

Enable / Disable JTAG mechanism

Node name	<code>\$bsp_path/jtag/enable</code>		
Description	Enable or disable JTAG mechanism		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	<code>enable</code>	Integer	0/1
Example	Enable JTAG mechanism: <code>echo 1 > \$bsp_path/jtag/enable</code>		

Set JTAG TCK pin

Node name	\$bsp_path/jtag/tck		
Description	Set JTAG TCK pin		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	tck	Integer	
Example	Set JTAG TCK pin: echo 1 > \$bsp_path/jtag/tck		

Set JTAG TDI pin

Node name	\$bsp_path/jtag/tdi		
Description	Set JTAG TDI pin		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	tdi	Integer	
Example	Set JTAG TDI pin: echo 1 > \$bsp_path/jtag/tdi		

Set JTAG TMS pin

Node name	\$bsp_path/jtag/tms		
Description	Set JTAG TMS pin		
Access	Write		
Release version	1.0		
Arguments	Name	Data type	Values
	tms	Integer	
Example	Set JTAG TMS pin: echo 1 > \$bsp_path/jtag/tms		

Get JTAG TDO pin

Node name	\$bsp_path/jtag/tdo		
Description	Get JTAG TDO pin		
Access	Read only		
Release version	1.0		
Arguments	Name	Data type	Values
	tdo	Integer	
Example	Get JTAG TDO pin: cat \$bsp_path/jtag/tdo		

Thermal Control

The thermal algorithm is described in a separate document - Thermal Monitoring for NVIDIA Systems with Third Party OS

Thermal Control Versions

The hw-management package supports two versions of thermal control:

TC v2.0 (Default)

- **Script:** `/usr/bin/hw_management_thermal_control.py`
- **Service:** `hw-management-tc.service`
- **Description:** Standard thermal control implementation for most systems

TC v2.5 (Advanced)

- **Script:** `/usr/bin/hw_management_thermal_control_2_5.py`
- **Service:** `hw-management-tc.service` (automatically selects version)
- **Description:** Enhanced thermal control with improved algorithms for newer systems
- **Supported Systems:** SN5610, SN5640, Q3401-RD

The system automatically selects the appropriate thermal control version based on the hardware platform.

Drivers

Watchdog

NVIDIA watchdog device is implemented in a programmable logic device. There are 2 types of HW watchdog implementations:

- Type 1 – actual HW timeout defined as a power of 2 msec. For example: Timeout 20 sec is

rounded up to 32768 msec. The maximum timeout period is 32 sec (32768 msec). Get time-left is not supported.

- Type 2 – actual HW timeout defined in seconds and is the same as user-defined timeout.

Maximum timeout is 255 sec. Get time-left is supported.

Type 1 HW watchdog implementation exists in old systems and all new systems have Type 2 HW watchdog. The two types of HW implementation also have a different register map.

NVIDIA systems can have 2 watchdogs: Main and auxiliary. Main and auxiliary watchdog devices can be enabled together on the same system. There are several actions that can be defined in the watchdog: System reset, start fans on full speed, and increase register counter. The last 2 actions are performed without a system reset. Actions without reset are provided for the auxiliary watchdog device, which is optional.

Watchdog can be started during a probe. In this case it is pinged by the watchdog core before the watchdog device is opened by the user space application.

Watchdog can be initialized in using a nowayout method. That is, once started it cannot be stopped. The mlx-wdt driver supports both HW watchdog implementations.

Watchdog driver is probed from the common mlx_platform driver. Mlx_platform driver provides an appropriate set of registers for NVIDIA watchdog device, identity name (mlx-wdt-main or mlx-wdt-aux), initial timeout, performed action in expiration and configuration flags.

Watchdog configuration flags: nowayout and start_at_boot. HW watchdog version: type1 or type2. The driver checks during initialization if the previous system reset was done by the watchdog. If yes, it makes a notification about this event.

Access to HW registers is performed through a generic regmap interface.